will omit this change, for simplicity. The combination of multiway branching and the TST representation by themselves is quite effective in many applications, but patricia-style collapse of one-way branching will further enhance performance when the keys are such that they are likely to match for long stretches (see Exercise 15.71).

Another easy improvement to TST-based search is to use a large explicit multiway node at the root. The simplest way to proceed is to keep a table of R TSTs: one for each possible value of the first letter in the keys. If R is not large, we might use the first two letters of the keys (and a table of size R^2). For this method to be effective, the leading digits of the keys must be well-distributed. The resulting hybrid search algorithm corresponds to the way that a human might search for names in a telephone book. The first step is a multiway decision ("Let's see, it starts with 'A' "), followed perhaps by some two-way decisions ("It's before 'Andrews,' but after 'Aitken"') followed by sequential character matching (" 'Algonquin,' ... No, 'Algorithms' isn't listed, because nothing starts with 'Algor'!").

Programs 15.10 and 15.11 are TST-based implementations of the symbol-table *search* and *insert* operations that use R-way branching at the root and that keep keys in leaves (so there is no one-way branching once the keys are distinguished). These programs are likely to be among the fastest available for searching with string keys. The underlying TST structure can also support a host of other operations.

In a symbol table that grows to be huge, we may want to adapt the branching factor to the table size. In Chapter 16, we shall see a systematic way to grow a multiway trie so that we can take advantage of multiway radix search for arbitrary file sizes.

Property 15.8 *A search or insertion in a TST with items in leaves (no one-way branching at the bottom) and R^t-way branching at the root requires roughly $\ln N - t \ln R$ byte accesses for N keys that are random bytestrings. The number of links required is R^t (for the root node) plus a small constant times N.*

These rough estimates follow immediately from Property 15.6. For the time cost, we assume that all but a constant number of the nodes on the search path (a few at the top) act as random BSTs on R character values, so we simply multiply the time cost by $\ln R$. For the space cost, we assume that the nodes on the first few levels are filled with R

Program 15.10 TST insertion for symbol-table ADT

This implementation of *insert* using TSTs keeps records in leaves, generalizing Program 15.3. If a search ends in a leaf, we create the internal nodes needed to distinguish the key found from the search key. We also improve on Program 15.8 by including R-way branching at the root node: Rather than using a single pointer head, we use an array heads of R links, indexed by the first digits of the keys. To initialize (code not shown) we set all R head links to NULL.

```
#define internal(A) ((A->d) != NULLdigit)
link NEWx(link h, int d)
  { link x = malloc(sizeof *x);
    x->item = NULLitem; x->d = d;
    x->l = NULL; x->m = h; x->r = NULL;
    return x;
  }
link split(link p, link q, int w)
  { int pd = digit(p->item, w),
        qd = digit(q->item, w);
    link t = NEW(NULLitem, qd);
    if (pd < qd) { t->m = q; t->l = NEWx(p, pd); }
    if (pd == qd) { t->m = split(p, q, w+1); }
    if (pd > qd) { t->m = q; t->r = NEWx(p, pd); }
    return t;
  }
link insertR(link h, Item item, int w)
  { Key v = key(item);
    int i = digit(v, w);
    if (h == NULL)
      return NEWx(NEW(item, NULLdigit), i);
    if (!internal(h))
      return split(NEW(item, NULLdigit), h, w);
    if (i < h->d) h->l = insertR(h->l, v, w);
    if (i == h->d) h->m = insertR(h->m, v, w+1);
    if (i > h->d) h->r = insertR(h->r, v, w);
    return h;
  }
void STinsert(Key key)
  { int i = digit(key, 0);
    heads[i] = insertR(heads[i], key, 1);
  }
```

Program 15.11 TST search for symbol-table ADT

This implementation of *search* for TSTs (built with Program 15.10) is like straight multiway-trie search, but we use only three, rather than R, links per node. We use the digits of the key to travel down the tree, ending either at a null link (search miss) or at a leaf that has a key that either is (search hit) or is not (search miss) equal to the search key.

```
Item searchR(link h, Key v, int w)
  { int i = digit(v, w);
    if (h == NULL) return NULLitem;
    if (internal(h))
      {
        if (i < h->d) return searchR(h->l, v, w);
        if (i == h->d) return searchR(h->m, v, w+1);
        if (i > h->d) return searchR(h->r, v, w);
      }
    if eq(v, key(h->item)) return h->item;
    return NULLitem;
  }
Item STsearch(Key v)
  { return searchR(heads[digit(v, 0)], v, 1); }
```

character values, and that the nodes on the bottom levels have only a constant number of character values. ▪

For example, if we have 1 billion random bytestring keys with $R = 256$, and we use a table of size $R^2 = 65536$ at the top, then a typical search will require about $\ln 10^9 - 2 \ln 256 \approx 20.7 - 11.1 = 9.6$ byte comparisons. Using the table at the top cuts the search cost by a factor of 2. If we have truly random keys, we can achieve this performance with more direct algorithms that use the leading bytes in the key and an existence table, in the manner discussed in Section 14.6. With TSTs, we can get the same kind of performance when keys have a less random structure.

It is instructive to compare TSTs without multiway branching at the root with standard BSTs, for random keys. Property 15.8 says that TST search will require about $\ln N$ *byte* comparisons, whereas standard BSTs require about $\ln N$ *key* comparisons. At the top of the BST, the key comparisons can be accomplished with just one byte

comparison, but at the bottom of the tree multiple byte comparisons may be needed to accomplish a key comparison. This performance difference is not dramatic. The reasons that TSTs are preferable to standard BSTs for string keys are that they provide a fast search miss; they adapt directly to multiway branching at the root; and (most important) they adapt well to bytestring keys that are *not* random, so no search takes longer than the length of a key in a TST.

Some applications may not benefit from the R-way branching at the root—for example, the keys in the library-call-number example of Figure 15.18 all begin with either `L` or `W`. Other applications may call for a higher branching factor at the root—for example, as just noted, if the keys were random integers, we would use as large a table as we could afford. We can use application-specific dependencies of this sort to tune the algorithm to peak performance, but we should not lose sight of the fact that one of the most attractive features of TSTs is that TSTs free us from having to worry about such application-specific dependencies, providing good performance without any tuning.

Perhaps the most important property of tries or TSTs with records in leaves is that their performance characteristics are *independent* of the key length. Thus, we can use them for arbitrarily long keys. In Section 15.5, we examine a particularly effective application of this kind.

Exercises

▷ 15.48 Draw the existence trie that results when you insert the words `now is the time for all good people to come the aid of their party` into an initially empty trie. Use 27-way branching.

▷ 15.49 Draw the existence TST that results when you insert the words `now is the time for all good people to come the aid of their party` into an initially empty TST.

▷ 15.50 Draw the 4-way trie that results when you insert items with the keys 01010011 00000111 00100001 01010001 11101100 00100001 10010101 0100-1010 into an initially empty trie, using 2-bit bytes.

▷ 15.51 Draw the TST that results when you insert items with the keys 0101-0011 00000111 00100001 01010001 11101100 00100001 10010101 01001010 into an initially empty TST, using 2-bit bytes.

▷ 15.52 Draw the TST that results when you insert items with the keys 0101-0011 00000111 00100001 01010001 11101100 00100001 10010101 01001010 into an initially empty TST, using 4-bit bytes.

○ 15.53 Draw the TST that results when you insert items with the library-call-number keys in Figure 15.18 into an initially empty TST.

○ 15.54 Modify our multiway-trie search and insertion implementation (Program 15.7) to work under the assumption that keys are (fixed-length) w-byte words (so no end-of-key indication is necessary).

○ 15.55 Modify our TST search and insertion implementation (Program 15.8) to work under the assumption that keys are (fixed-length) w-byte words (so no end-of-key indication is necessary).

15.56 Run empirical studies to compare the time and space requirements of an 8-way trie built with random integers using 3-bit bytes, a 4-way trie built with random integers using 2-bit bytes, and a binary trie built from the same keys, for $N = 10^3$, 10^4, 10^5, and 10^6 (see Exercise 15.14).

15.57 Modify Program 15.9 such that it visits, in the same manner as *sort*, all the nodes that match the search key.

○ 15.58 Write a function that prints all the keys in a TST that differ from the search key in at most k positions, for a given integer k.

• 15.59 Give a full characterization of the worst-case internal path length of an R-way trie with N distinct w-bit keys.

• 15.60 Implement the *sort*, *delete*, *select*, and *join* operations for a multiway-trie–based symbol table.

• 15.61 Implement the *sort*, *delete*, *select*, and *join* operations for a TST-based symbol table.

▷ 15.62 Write a program that prints out all keys in an R-way trie that have the same initial t bytes as a given search key.

• 15.63 Modify our multiway-trie search and insertion implementation (Program 15.7) to eliminate one-way branching in the way that we did for patricia tries.

• 15.64 Modify our TST search and insertion implementation (Program 15.8) to eliminate one-way branching in the way that we did for patricia tries.

15.65 Write a program to balance the BSTs that represent the internal nodes of a TST (rearrange them such that all their external nodes are on one of two levels).

15.66 Write a version of *insert* for TSTs that maintains a balanced-tree representation of all the internal nodes (see Exercise 15.65).

• 15.67 Give a full characterization of the worst-case internal path length of a TST with N distinct w-bit keys.

15.68 Write a program that generates random 80-byte string keys (see Exercise 10.19). Use this key generator to build a 256-way trie with N random keys, for $N = 10^3$, 10^4, 10^5, and 10^6, using *search*, then *insert* on search miss.

Instrument your program to print out the total number of nodes in each trie and the total amount of time taken to build each trie.

15.69 Answer Exercise 15.68 for TSTs. Compare your performance results with those for tries.

15.70 Write a key generator that generates keys by shuffling a random 80-byte sequence (see Exercise 10.21). Use this key generator to build a 256-way trie with N random keys, for $N = 10^3$, 10^4, 10^5, and 10^6, using *search*, then *insert* on search miss. Compare your performance results with those for the random case (see Exercise 15.68).

○ 15.71 Write a key generator that generates 30-byte random strings made up of three fields: a 4-byte field with one of a set of 10 given strings; a 10-byte field with one of a set of 50 given strings; a 1-byte field with one of two given values; and a 15-byte field with random left-justified strings of letters equally likely to be four through 15 characters long (see Exercise 10.23). Use this key generator to build a 256-way trie with N random keys, for $N = 10^3$, 10^4, 10^5, and 10^6, using *search*, then *insert* on search miss. Instrument your program to print out the total number of nodes in each trie and the total amount of time taken to build each trie. Compare your performance results with those for the random case (see Exercise 15.68).

15.72 Answer Exercise 15.71 for TSTs. Compare your performance results with those for tries.

15.73 Develop an implementation of *search* and *insert* for bytestring keys using multiway *digital* search trees.

▷ 15.74 Draw the 27-way DST (see Exercise 15.73) that results when you insert items with the keys `now is the time for all good people to come the aid of their party` into an initially empty DST.

• 15.75 Develop an implementation of multiway-trie search and insertion using linked lists to represent the trie nodes (as opposed to the BST representation that we use for TSTs). Run empirical studies to determine whether it is more efficient to use ordered or unordered lists, and to compare your implementation with a TST-based implementation.

15.5 Text-String–Index Algorithms

In Section 12.7, we considered the process of building a *string index*, and we used a binary search tree with string pointers to provide the capability to determine whether or not a given key string appears in a huge text. In this section, we look at more sophisticated algorithms using multiway tries, but starting from the same point of departure. We consider each position in the text to be the beginning of a string key that runs all the way to the end of the text and build a symbol table

with these keys, using string pointers. The keys are all different (for example, they are of different lengths), and most of them are extremely long. The purpose of a search is to determine whether or not a given search key is a prefix of one of the keys in the index, which is equivalent to discovering whether the search key appears somewhere in the text string.

A search tree that is built from keys defined by string pointers into a text string is called a *suffix tree*. We could use any algorithm that can admit variable-length keys. Trie-based methods are particularly suitable, because (except for the trie methods that do one-way branching at the tails of keys) their running time does not depend on the key length, but rather depends on only the number of digits required to distinguish among the keys. This characteristic lies in direct contrast to, for example, hashing algorithms, which do not apply immediately to this problem because their running time is proportional to the key length.

Figure 15.20 gives examples of string indexes built with BSTs, patricia, and TSTs (with leaves). These indexes use just the keys starting at word boundaries; an index starting at character boundaries would provide a more complete index, but would use significantly more space.

Strictly speaking, even a random string text does not give rise to a random set of keys in the corresponding index (because the keys are not independent). However, we rarely work with random texts in practical indexing applications, and this analytic discrepancy will not stop us from taking advantage of the fast indexing implementations that are possible with radix methods. We refrain from discussing the detailed performance characteristics when we use each of the algorithms to build a string index, because many of the same tradeoffs that we have discussed for general symbol tables with string keys also hold for the string-index problem.

For a typical text, standard BSTs would be the first implementation that we might choose, because they are simple to implement (see Program 12.10). For typical applications, this solution is likely to provide good performance. One byproduct of the interdependence of the keys—particularly when we are building a string index for each character position—is that the worst case for BSTs is not a particular

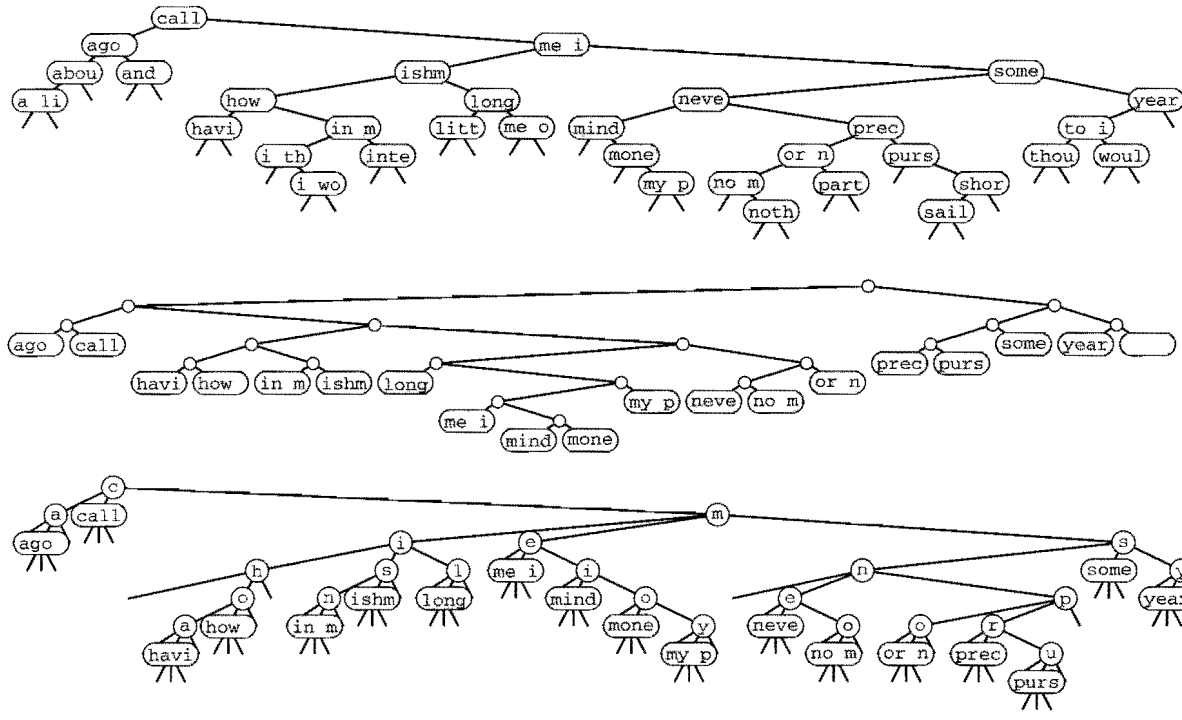


Figure 15.20
Text-string index examples

These diagrams show text-string indexes built from the text `call me ishmael some years ago never mind how long precisely` *... using a BST* (top)*, a patricia trie* (center)*, and a TST* (bottom)*. Nodes with string pointers are depicted with the first four characters at the point referenced by the pointer.*

concern for huge texts, since unbalanced BSTs occur with only bizarre constructions.

Patricia was originally designed for the string-index application. To use Programs 15.5 and 15.4, we need only to provide an implementation of `bit` that, given a string pointer and an integer i, returns the ith bit of the string (see Exercise 15.81). In practice, the height of a patricia trie that implements a text string index will be logarithmic. Moreover, a patricia trie will provide fast search implementations for misses because we do not need to examine all the bytes of the key.

TSTs afford several of the performance advantages of patricia, are simple to implement, and take advantage of built-in byte-access operations that are typically found on modern machines. They also are amenable to simple implementations, such as Program 15.9, that can solve search problems more complicated than fully matching a search key. To use TSTs to build a string index, we need to remove the code that handles ends of keys in the data structure, since we are

guaranteed that no string is a prefix of another, and thus we never will be comparing strings to their ends. This modification includes changing the definition of eq in the item-type interface to regard two strings as equal if one is a prefix of the other, as we did in Section 12.7, since we will be comparing a (short) search key against a (long) text string, starting at some position in the text string. A third change that is convenient is to keep string pointers in each node, rather than characters, so that every node in the tree refers to a position in the text string (the position in the text string following the first occurrence of the character string defined by the characters on equal branches from the root to that node). Implementing these changes is an interesting and informative exercise that leads to a flexible and efficient text-string–index implementation (see Exercise 15.80).

Despite all the advantages that we have been discussing, there is an important fact that we are overlooking when considering the use of BSTs, patricia tries, or TSTs for typical text indexing applications: the text itself is usually fixed, so we do not need to support the dynamic *insert* operations that we have become accustomed to supporting. That is, we typically build the index once, then use it for a huge number of searches, without ever changing it. Therefore, we may not need dynamic data structures like BSTs, patricia tries or TSTs at all. The basic algorithm that is appropriate for handling this situation is *binary search*, with string pointers (see Section 12.4). The index is a set of string pointers; index construction is a string pointer sort. The primary advantage of using binary search over a dynamic data structure is the space savings. To index a text string at N positions using binary search, we need just N string pointers; in contrast, to index a string at N positions using a tree-based method, we need at least $3N$ pointers (one string pointer, to the text, and two links). Text indexes are typically huge, so binary search might be preferred because it provides guaranteed logarithmic search time but uses one-third the amount of memory used by tree-based methods. If sufficient memory space is available, however, TSTs will lead to a faster *search* for many applications because it moves through the key without retracing its steps, and binary search does not do so.

If we have a huge text but plan to perform only a small number of searches, then building a full index is not likely to be justified. In Part 5, we consider the *string-search* problem, where we want to determine

quickly whether a given text contains a given search key, without any preprocessing. We shall also consider a number of string-search problems that are between the two extremes of doing no preprocessing and building a full index for a huge text.

Exercises

▷ 15.76 Draw the 26-way DST that results when you build a text-string index from the words `now is the time for all good people to come the aid of their party`.

▷ 15.77 Draw the 26-way trie that results when you build a text-string index from the words `now is the time for all good people to come the aid of their party`.

▷ 15.78 Draw the TST that results when you build a text-string index from the words `now is the time for all good people to come the aid of their party`, in the style of Figure 15.20.

▷ 15.79 Draw the TST that results when you build a text-string index from the words `now is the time for all good people to come the aid of their party`, using the implementation described in the text, where the TST contains string pointers at every node.

○ 15.80 Modify the TST search and insertion implementations in Programs 15.10 and 15.11 to provide a TST-based string index.

○ 15.81 Implement an interface that allows patricia to process C string keys (that is, arrays of characters) as though they were bitstrings.

○ 15.82 Draw the patricia trie that results when you build a text string index from the words `now is the time for all good people to come the aid of their party`, using a 5-bit binary coding with the ith letter in the alphabet represented by the binary representation of i.

15.83 Explain why the idea of improving binary search using the same basic principle on which TSTs are based (comparing characters rather than strings) is not effective.

15.84 Find a large (at least 10^6 bytes) text file on your system, and compare the height and internal path length of a standard BST, patricia trie, and TST, when you use these methods to build an index from that file.

15.85 Run empirical studies to compare the height and internal path length of a standard BST, patricia trie, and TST, when you use these methods to build an index from a text string consisting of N random characters from a 32-character alphabet, for $N = 10^3$, 10^4, 10^5, and 10^6.

○ 15.86 Write an efficient program to determine the longest repeated sequence in a huge text string.

○ 15.87 Write an efficient program to determine the 10-character sequence that occurs most frequently in a huge text string.

• 15.88 Build a string index that supports an operation that returns the number of occurrences of its argument in the indexed text, and supports, in the same manner as *sort*, a *search* operation that visits all the text positions that match the search key.

○ 15.89 Describe a text string of N characters for which a TST-based string index will perform particularly badly. Estimate the cost of building an index for the same string with a BST.

15.90 Suppose that we want to build an index for a random N-bit string, for bit positions that are a multiple of 16. Run empirical studies to determine which of the bytesizes 1, 2, 4, 8, or 16 leads to the lowest running times to construct a TST-based index, for $N = 10^3$, 10^4, 10^5, and 10^6.

CHAPTER SIXTEEN

External Searching

SEARCH ALGORITHMS THAT are appropriate for accessing items from huge files are of immense practical importance. Searching is the fundamental operation on huge data sets, and it certainly consumes a significant fraction of the resources used in many computing environments. With the advent of world wide networking, we have the ability to gather almost all the information that is possibly relevant to a task—our challenge is to be able to search through it efficiently. In this chapter, we discuss basic underlying mechanisms that can support efficient search in symbol tables that are as large as we can imagine.

Like those in Chapter 11, the algorithms that we consider in this chapter are relevant to numerous different types of hardware and software environments. Accordingly, we tend to think about the algorithms at a level more abstract than that of the C programs that we have been considering. However, the algorithms that we shall consider also directly generalize familiar searching methods, and are conveniently expressed as C programs that are useful in many situations. We will proceed in a manner different from that in Chapter 11: We develop specific implementations in detail, consider their essential performance characteristics, and then discuss ways in which the underlying algorithms might prove useful under situations that might arise in practice. Taken literally, the title of this chapter is a misnomer, since we will present the algorithms as C programs that we could substitute interchangeably for the other symbol-table implementations that we have considered in Chapters 12 through 15. As such, they are not "external" methods at all. However, they are built in accordance with

a simple abstract model, which makes them precise specifications of how to build searching methods for specific external devices.

Detailed abstract models are less useful than they were for sorting because the costs involved are so low for many important applications. We shall be concerned mainly with methods for searching in huge files stored on any external device where we have fast access to arbitrary blocks of data, such as a disk. For tapelike devices, where only sequential access is allowed (the model that we considered in Chapter 11), searching degenerates to the trivial (and slow) method of starting at the beginning and reading until completion of the search. For disklike devices, we can do much better: Remarkably, the methods that we shall study can support *search* and *insert* operations on symbol tables containing billions or trillions of items using only three or four references to blocks of data on disk. System parameters such as block size and the ratio of the cost of accessing a new block to the cost of accessing the items within a block affect performance, but the methods are relatively insensitive to the values of these parameters (within the ranges of values that we are likely to encounter in practice). Moreover, the most important steps that we must take to adapt the methods to particular practical situations are straightforward.

Searching is a fundamental operation for disk devices. Files are typically organized to take advantage of particular device characteristics to make access to information as efficient as possible. In short, it is safe to assume that the devices that we use to store huge amounts of information are built to support efficient and straightforward implementations of *search*. In this chapter, we consider algorithms built at a level of abstraction slightly higher than that of the basic operations provided by disk hardware, which can support *insert* and other dynamic symbol-table operations. These methods afford the same kinds of advantages over the straightforward methods that BSTs and hash tables offer over binary search and sequential search.

In many computing environments, we can address a huge *virtual memory* directly, and can rely on the system to find efficient ways to handle any program's requests for data. The algorithms that we consider also can be effective solutions to the symbol-table implementation problem in such environments.

A collection of information to be processed with a computer is called a *database*. A great deal of study has gone into methods of

building, maintaining and using databases. Much of this work has been in the development of abstract models and implementations to support *search* operations with criteria more complex than the simple "match a single key" criterion that we have been considering. In a database, searches might be based on criteria for partial matches perhaps including multiple keys, and might be expected to return a large number of items. We touch on methods of this type in Parts 5 and 6. General search requests are sufficiently complicated that it is not atypical for us to do a sequential search over the entire database, testing each item to see if it meets the criteria. Still, fast search for tiny bits of data matching specific criteria in a huge file is an essential capability in any database system, and many modern databases are built on the mechanisms that we describe in this chapter.

16.1 Rules of the Game

As we did in Chapter 11, we make the basic assumption that sequential access to data is far less expensive than nonsequential access. Our model will be to consider whatever memory facility that we use to implement the symbol table as divided up into *pages*: Contiguous blocks of information that can be accessed efficiently by the disk hardware. Each page will hold many items; our task is to organize the items within the pages such that we can access any item by reading only a few pages. We assume that the I/O time required to read a page completely dominates the processing time required to access specific items or to do any other computing involving that page. This model is oversimplified in many ways, but it retains enough of the characteristics of external storage devices to allow us to consider fundamental methods.

Definition 16.1 *A* **page** *is a contiguous block of data. A* **probe** *is the first access to a page.*

We are interested in symbol-table implementations that use a small number of probes. We avoid making specific assumptions about the page size and about the ratio of the time required for a probe to the time required, subsequently, to access items within the block. We expect these values to be on the order of 100 or 1000; we do not need

to be more precise because the algorithms are not highly sensitive to these values.

This model is directly relevant, for example, in a file system in which files comprise blocks with unique identifiers and in which the purpose is to support efficient access, insertion, and deletion based on that identifier. A certain number of items fit on a block, and the cost of processing items within a block is insignificant compared to the cost of reading the block.

This model is also directly relevant in a virtual-memory system, where we simply refer directly to a huge amount of memory, and rely on the system to keep the information that we use most often in fast storage (such as internal memory) and the information that we use infrequently in slow storage (such as a disk). Many computer systems have sophisticated paging mechanisms, which implement virtual memory by keeping recently used pages in a *cache* that can be accessed quickly. Paging systems are based on the same abstraction that we are considering: They divide the disk into blocks, and assume that the cost of accessing a block for the first time far exceeds the cost of accessing data within the block.

Our abstract notion of a page typically will correspond precisely to a block in a file system or to a page in a virtual-memory system. For simplicity, we generally assume this correspondence when considering the algorithms. For specific applications, we might have multiple pages per block or multiple blocks per page for system- or application-dependent reasons; such details do not diminish the effectiveness of the algorithms, and thus underscore the utility of working at an abstract level.

We manipulate pages, references to pages, and items with keys. For a huge database, the most important problem to consider now is to maintain an *index* to the data. That is, as discussed briefly in Section 12.7, we assume that the items constituting our symbol table are stored in some static form somewhere, and that our task is to build a data structure with keys and references to items that allows us to produce quickly a reference to a given item. For example, a telephone-company might have customer information stored in a huge static database, with several indexes on the database, perhaps using different keys, for monthly billing, daily transaction processing, periodic solicitation, and so forth. For huge data sets, indexes are of

critical importance: We generally do not make copies of the basic data, not only because we may not be able to afford the extra space, but also because we want to avoid the problems associated with maintaining the integrity of the data when we have multiple copies.

Accordingly, we generally assume that each item is a *reference* to the actual data, which may be a page address or some more complex interface to a database. For simplicity, we do not keep copies of items in our data structures, but we do keep copies of keys—an approach that is often practical. Also, for simplicity in describing the algorithms, we do not use an abstract interface for item and page references—instead, we just use pointers. Thus, we can use our implementations directly in a virtual-memory environment, but have to convert the pointers and pointer access into more complex mechanisms to make them true external sorting methods.

We shall consider algorithms that, for a broad range of values of the two main parameters (block size and relative access time), implement *search*, *insert*, and other operations in a fully dynamic symbol table using only a few probes per operation. In the typical case where we perform huge numbers of operations, careful tuning might be effective. For example, if we can reduce the typical search cost from three probes to two probes, we might improve system performance by 50 percent! However, we will not consider such tuning here; its effectiveness is strongly system- and application-dependent.

On ancient machines, external storage devices were complex contraptions that not only were big and slow, but also did not hold much information. Accordingly, it was important to work to overcome their limitations, and early programming lore is filled with tales of external file access programs that were timed perfectly to grab data off a rotating disk or drum and otherwise to minimize the amount of physical movement required to access data. Early lore is also filled with tales of spectacular failures of such attempts, where slight miscalculations made the process much slower than a naive implementation would have been. By contrast, modern storage devices not only are tiny and extremely fast, but also hold huge amounts of information; so we generally do not need to address such problems. Indeed, in modern programming environments, we tend to shy away from dependencies on the properties of specific physical devices—it is generally more important that our programs be effective on a variety of machines

(including those to be developed in the future) than that they achieve peak performance for a particular device.

For long-lived databases, there are numerous important implementation issues surrounding the general goals of maintaining the integrity of the data and providing flexible and reliable access. We do not address such issues here. For such applications, we may view the methods that we consider as the underlying algorithms that will ultimately ensure good performance, and as a starting point in the system design.

16.2 Indexed Sequential Access

A straightforward approach to building an index is to keep an array with keys and item references, in order of the keys, then to use binary search (see Section 12.4) to implement *search*. For N items, this method would require $\lg N$ probes—even for a huge file. Our basic model leads us immediately to consider two modifications to this simple method. First, the index itself is huge and will not fit on a single page, in general. Since we can access pages only through page references, we can build, instead, an explicit fully balanced binary tree with keys and page pointers in internal nodes, and with keys and item pointers in external nodes. Second, the cost of accessing M table entries is the same as the the cost of accessing 2, so we can use an M-ary tree for about the same cost per node as a binary tree. This improvement reduces the number of probes to be proportional to about $\log_M N$. As we saw in Chapters 10 and 15, we can regard this quantity to be constant for practical purposes. For example, if M is 1000, then $\log_M N$ is less than 5 if N is less than 1 trillion.

Figure 16.1 gives a sample set of keys, and Figure 16.2 depicts an example of such a tree structure for those keys. We need to use relatively small values of M and N to keep our examples manageable; nevertheless, they illustrate that the trees for large M will be flat.

The tree depicted in Figure 16.2 is an abstract device-independent representation of an index that is similar to many other data structures that we have considered. Note that, in addition, it is not far removed from device-*dependent* indexes that might be found in low-level disk access software. For example, some early systems used a two-level scheme, where the bottom level corresponded to the items on the

706	111000110
176	001111110
601	110000001
153	001101011
513	101001011
773	111111011
742	111100010
373	011111011
524	101010100
766	111110110
275	010111101
737	111011111
574	101111100
434	100011100
641	110100001
207	010000111
001	000000001
277	010111111
061	000110001
736	111011110
526	101010110
562	101110010
017	000001111
107	001000111
147	001100111

Figure 16.1
Binary representation of octal keys

The keys (left) *that we use in the examples in this chapter are 3-digit octal numbers, which we also interpret as 9-bit binary values* (right).

pages for a particular disk device, and the second level corresponded to a master index to the individual devices. In such systems, the master index was kept in main memory, so accessing an item with such an index required two disk accesses: one to get the index, and one to get the page containing the item. As disk capacity increases, so increases the size of the index, and several pages might be required to store the index, eventually leading to a hierarchical scheme like the one depicted in Figure 16.2. We shall continue working with an abstract representation, secure in the knowledge that it can be implemented directly with typical low-level system hardware and software.

Many modern systems use a similar tree structure to organize huge files as a sequence of disk pages. Such trees contain no keys, but they can efficiently support the commonly used operations of accessing the file in sequential order, and, if each node contains a count of its tree size, of finding the page containing the kth item in the file.

Historically, because it combines a sequential key organization with indexed access, the indexing method depicted in Figure 16.2 is called *indexed sequential access*. It is the method of choice for applications in which changes to the database are rare. We sometimes refer to the index itself as a *directory*. The disadvantage of using indexed sequential access is that modifying the directory is an expensive operation. For example, adding a single key can require rebuilding virtually the whole database, with new positions for many of the keys and new values for the indexes. To combat this defect and to provide for modest growth, early systems provided for overflow pages on disks and overflow space in pages, but such techniques ultimately were not very effective in dynamic situations (see Exercise 16.3). The methods that we consider in Sections 16.3 and 16.4 provide systematic and efficient alternatives to such ad hoc schemes.

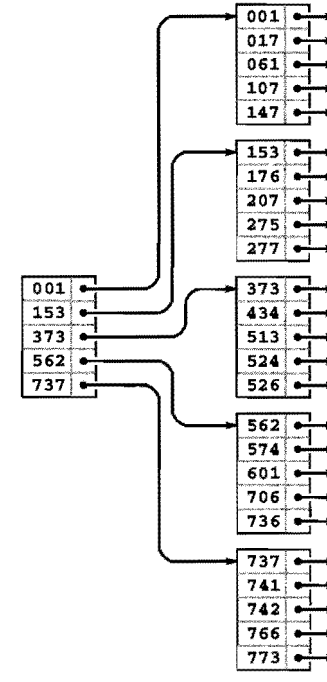


Figure 16.2
Indexed sequential file structure

In a sequential index, we keep the keys in sequential order in full pages (right)*, with an index directing us to the smallest key in each page* (left)*. To add a key, we need to rebuild the data structure.*

Property 16.1 *A search in an indexed sequential file requires only a constant number of probes, but an insertion can involve rebuilding the entire index.*

We use the term *constant* loosely here (and throughout this chapter) to refer to a quantity that is proportional to $\log_M N$ for large M. As we have discussed, this usage is justified for practical file sizes. Figure 16.3 gives more examples. Even if we were to have a 128-bit search key, capable of specifying the impossibly large number of 2^{128}

different items, we could find an item with a given key in 13 probes, with 1000-way branching. ■

We will not consider implementations that search and construct indexes of this type, because they are special cases of the more general mechanisms that we consider in Section 16.3 (see Exercise 16.17 and Program 16.2).

Exercises

▷ 16.1 Tabulate the values of $\log_M N$, for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

▷ 16.2 Draw an indexed sequential file structure for the keys 516, 177, 143, 632, 572, 161, 774, 470, 411, 706, 461, 612, 761, 474, 774, 635, 343, 461, 351, 430, 664, 127, 345, 171, and 357 , for $M = 5$ and $M = 6$.

○ 16.3 Suppose that we build an indexed sequential file structure for N items in pages of capacity M, but leave k empty spaces in each page for expansion. Give a formula for the number of probes needed for a search, as a function of N, M, and k. Use the formula to determine the number of probes needed for a search when $k = M/10$, for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

○ 16.4 Suppose that the cost of a probe is about α time units, and that the average cost of finding an item in a page is about βM time units. Find the value of M that minimizes the cost for a search in an indexed sequential file structure, for $\alpha/\beta = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

10^5	words in dictionary
10^6	words in *Moby Dick*
10^9	Social Security numbers
10^{12}	phone numbers in the world
10^{15}	people who ever lived
10^{20}	grains of sand on beach at Coney Island
10^{25}	bits of memory ever manufactured
10^{79}	electrons in universe

Figure 16.3
Examples of data set sizes

These generous upper bounds indicate that we can assume safely, for practical purposes, that we will never have a symbol table with more than 10^{30} items. Even in such an unrealistically huge database, we could find an item with a given key with less than 10 probes, if we did 1000-way branching. Even if we somehow found a way to store information on each electron in the universe, 1000-way branching would give us access to any particular item with less than 27 probes.

16.3 B Trees

To build search structures that can be effective in a dynamic situation, we build multiway trees, but we relax the restriction that every node must have *exactly* M entries. Instead, we insist that every node must have *at most* M entries, so that they will fit on a page, but we allow nodes to have fewer entries. To be sure that nodes have a sufficient number of entries to provide the branching that we need to keep search paths short, we also insist that all nodes have *at least* (say) $M/2$ entries, except possibly the root, which must have a least one entry (two links). The reason for the exception at the root will become clear when we consider the construction algorithm in detail. Such trees were named *B trees* by Bayer and McCreight, who, in 1970, were the first researchers to consider the use of multiway balanced trees for external searching. Many people reserve the term *B tree* to describe the exact data structure

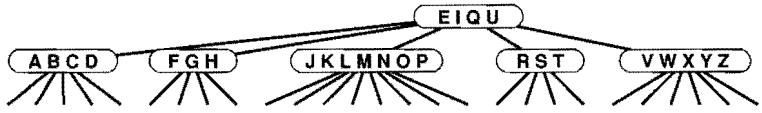


Figure 16.4
A 4-5-6-7-8 tree

This figure depicts a generalization of 2-3-4 trees built from nodes with 4 through 8 links (and 3 through 7 keys, respectively). As with 2-3-4 trees, we keep the height constant by splitting 8-nodes when encountering them, with either a top-down or a bottom-up insertion algorithm. For example, to insert another J *into this tree, we would first split the 8-node into two 4-nodes, then insert the* M *into the root, converting it into a 6-node. When the root splits, we have no choice but to create a new root that is a 2-node, so the root node is excused from the constraint that nodes must have at least four links.*

built by the algorithm suggested by Bayer and McCreight; we use it as a generic term to refer to a family of related algorithms.

We have already seen a B-tree implementation: In Definitions 13.1 and 13.2, we see that B trees of order 4, where each node has at most four links and at least two links, are none other than the balanced 2-3-4 trees of Chapter 13. Indeed, the underlying abstraction generalizes in a straightforward manner, and we can implement B trees by generalizing the top-down 2-3-4 tree implementations in Section 13.4. However, the various differences between external and internal searching that we discussed in Section 16.1 lead us to a number of different implementation decisions. In this section, we consider an implementation that

- Generalizes 2-3-4 trees to trees with between $M/2$ and M nodes
- Represents multiway nodes with an array of items and links
- Implements an index instead of a search structure containing the items
- Splits from the bottom up
- Separates the index from the items

The final two properties in this list are not essential, but are convenient in many situations and are normally found in B tree implementations.

Figure 16.4 illustrates an abstract 4-5-6-7-8 tree, which generalizes the 2-3-4 tree that we considered in Section 13.3. The generalization is straightforward: 4-nodes have three keys and four links, 5-nodes have four keys and five links, and so forth, with one link for each possible interval between keys. To search, we start at the root and move from one node to the next by finding the proper interval for the search key in the current node and then exiting through the corresponding link to get to the next node. We terminate the search with a search hit if we find the search key in any node that we touch; we terminate with a search miss if we reach the bottom of the tree without a hit. As we can in top-down 2-3-4 trees, we can insert a new key at the bottom of the tree after a search if, on the way down the tree, we split nodes that are full: If the root is an 8-node, we split

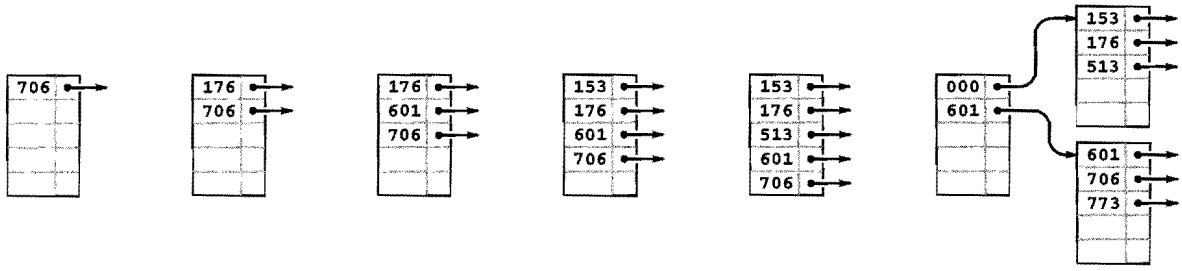


Figure 16.5
B tree construction, part 1

This example shows six insertions into an initially empty B tree built with pages that can hold five keys and links, using keys that are 3-digit octal numbers (9-bit binary numbers). We keep the keys in order in the pages. The sixth insertion causes a split into two external nodes with three keys each and an internal node that serves as an index: Its first pointer points to the page containing all keys greater than or equal to 000 but less than 601, and its second pointer points to the page containing all keys greater than or equal to 601.

it into a 2-node connected to two 4-nodes; then, any time we see a k-node attached to an 8-node, we replace it by a $(k+1)$-node attached to two 4-nodes. This policy guarantees that we have room to insert the new node when we reach the bottom.

Alternatively, as discussed for 2-3-4 trees in Section 13.3, we can split from the bottom up: We insert by searching and putting the new key in the bottom node, unless that node is a 8-node, in which case we split it into two 4-nodes and insert the middle key and the links to the two new nodes into its parent, working up the tree until encountering an ancestor that is not a 8-node.

Replacing 4 by $M/2$ and 8 by M in descriptions in the previous two paragraphs converts them into descriptions of search and insert for $M/2$-...-M trees, for any positive even integer M, even 2 (see Exercise 16.9).

Definition 16.2 *A* **B tree** *of order M is a tree that either is empty or comprises* **k-nodes**, *with $k-1$ keys and k links to trees representing each of the k intervals delimited by the keys, and has the following structural properties: k must be between 2 and M at the root and between $M/2$ and M at every other node; and all links to empty trees must be at the same distance from the root.*

B tree algorithms are built upon this basic set of abstractions. As in Chapter 13, we have a great deal of freedom in choosing concrete representations for such trees. For example, we can use an extended red-black representation (see Exercise 13.69). For external searching, we use the even more straightforward ordered-array representation, taking M to be sufficiently large that M-nodes fill a page. The branching factor is at least $M/2$, so the number of probes for any search or insert is effectively constant, as discussed following Property 16.1.

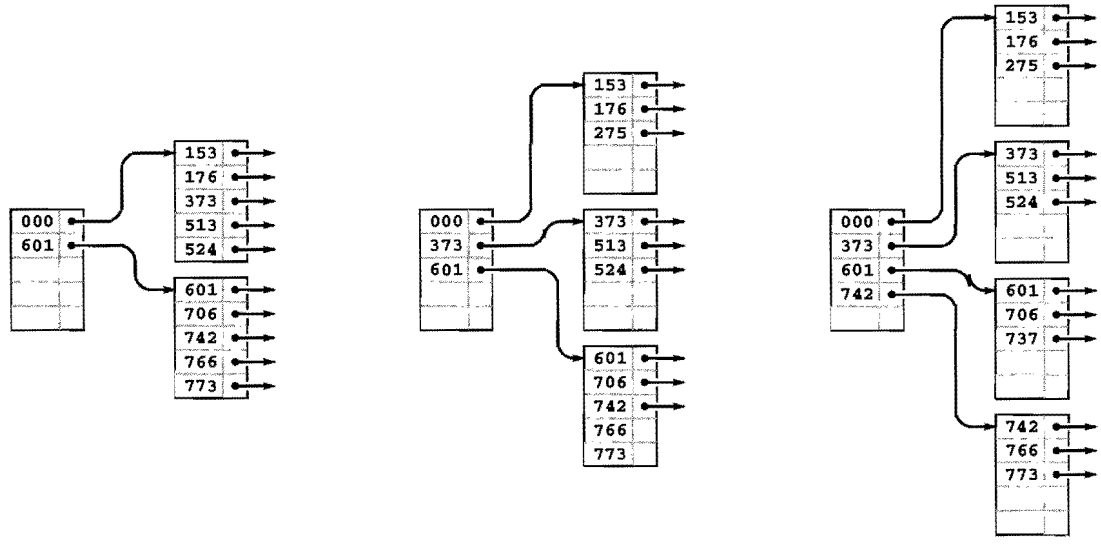


Figure 16.6
B tree construction, part 2

After we insert the four keys 742, 373, 524, *and* 766 *into the rightmost B tree in Figure 16.5, both of the external pages are full* (left). *Then, when we insert* 275, *the first page splits, sending a link to the new page (along with its smallest key* 373*) up to the index* (center); *when we then insert* 737, *the page at the bottom splits, again sending a link to the new page up to the index* (right).

Instead of implementing the method just described, we consider a variant that generalizes the standard index that we considered in Section 16.1. We keep keys with item references in *external pages* at the bottom of the tree, and copies of keys with page references in *internal pages*. We insert new items at the bottom, and then use the basic $M/2$-...-M tree abstraction. When a page has M entries, we split it into two pages with $M/2$ pages each, and insert a reference to the new page into its parent. When the root splits, we make a new root with two children, thus increasing the height of the tree by 1.

Figures 16.5 through 16.7 show the B tree that we build by inserting the keys in Figure 16.1 (in the order given) into an initially empty tree, with $M = 5$. Doing insertions involves simply adding an item to a page, but we can look at the final tree structure to determine the significant events that occurred during its construction. It has seven external pages, so there must have been six external node splits, and it is of height 3, so the root of the tree must have split twice. These events are described in the commentary that accompanies the figures.

Program 16.1 gives the type definitions for nodes and the initialization code for our B-tree implementation. It is similar to several other tree-search implementations that we have examined, in Chapters 13 and 15. The chief added wrinkle is that we use the C `union` construct to allow us to define slightly different external and internal

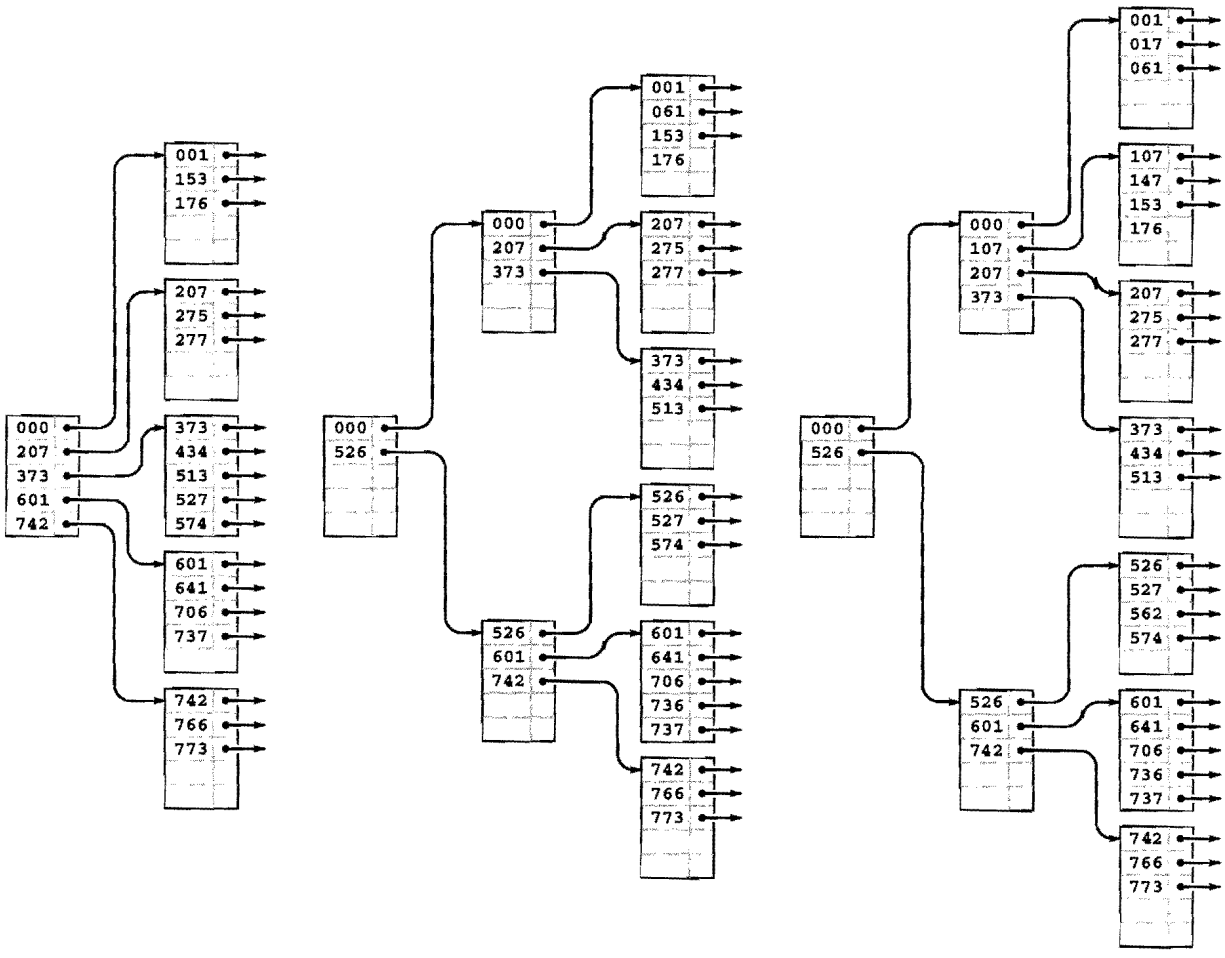


Figure 16.7
B tree construction, part 3

Continuing our example, we insert the 13 keys 574, 434, 641, 207, 001, 277, 061, 736, 526, 562, 017, 107, *and* 147 *into the rightmost B tree in Figure 16.6. Node splits occur when we insert* 277 (left), 526 (center), *and* 107 (right). *The node split caused by inserting* 526 *also causes the index page to split, and increases the height of the tree by one.*

nodes with the same structure (and the same type of link): Each node consists of an array of keys with associated links (in internal nodes) or items (in external nodes), and a count giving the number of active entries.

With these definitions and the example trees that we just considered, the code for *search* that is given in Program 16.2 is straightforward. For external nodes, we scan through the array of nodes to look for a key matching the search key, returning the associated item if we succeed and a null item if we do not. For internal nodes, we scan

Program 16.1 B-tree definitions and initialization

Each node in a B tree contains an array and a count of the number of active entries in the array. In internal nodes, each array entry is a key and a link to a node; in external nodes, each array entry is a key and an item. The C union construct allows us to specify these options in a single declaration.

We initialize new nodes to be empty (count field set to 0), with a sentinel key in array entry 0. An empty B tree is a link to an empty node. Also, we maintain variables to track the number of items in the tree and the height of the tree, both initialized to 0.

```
typedef struct STnode* link;
typedef struct
  { Key key; union { link next; Item item; } ref; }
entry;
struct STnode { entry b[M]; int m; };
static link head;
static int H, N;
link NEW()
  { link x = malloc(sizeof *x);
    x->m = 0;
    return x;
  }
void STinit(int maxN)
  { head = NEW(); H = 0; N = 0; }
```

through the array of nodes to find the link to the unique subtree that could contain the search key.

Program 16.3 is an implementation of *insert* for B trees; it too uses the recursive approach that we have taken for numerous other search-tree implementations in Chapters 13 and 15. It is a bottom-up implementation because we check for node splitting *after* the recursive call, so the first node split is an external node. The split requires that we pass up a new link to the parent of the split node, which in turn might need to split and pass up a link to its parent, and so forth, perhaps all the way up to the root of the tree (when the root splits, we create a new root, with two children). By contrast, the 2-3-4–tree implementation of Program 13.6 checks for splits *before* the recursive call, so we do splitting on the way down the tree. We could also use a

Program 16.2 B-tree search

This implementation of *search* for B trees is based on a recursive function, as usual. For internal nodes (positive height), we scan to find the first key larger than the search key, and do a recursive call on the subtree referenced by the previous link. For external nodes (height 0), we scan to see whether or not there is an item with key equal to the search key.

```
Item searchR(link h, Key v, int H)
  { int j;
    if (H == 0)
      for (j = 0; j < h->m; j++)
        if (eq(v, h->b[j].key))
          return h->b[j].ref.item;
    if (H != 0)
      for (j = 0; j < h->m; j++)
        if ((j+1 == h->m) || less(v, h->b[j+1].key))
          return searchR(h->b[j].ref.next, v, H-1);
    return NULLitem;
  }
Item STsearch(Key v)
  { return searchR(head, v, H); }
```

top-down approach for B trees (see Exercise 16.10). This distinction between top-down versus bottom-up approaches is unimportant in many B tree applications, because the trees are so flat.

The node-splitting code is given in Program 16.4. In the code, we use an even value for the variable M, and we allow only $M - 1$ items per node in the tree. This policy allows us to insert the Mth item into a node *before* splitting that node, and simplifies the code considerably without having much effect on the cost (see Exercises 16.20 and 16.21). For simplicity, we use the upper bound of M items per node in the analytic results later in this section; the actual differences are minute. In a top-down implementation, we would not have to resort to this technique, because the convenience of being sure that there is always room to insert a new key in each node comes automatically.

Property 16.2 *A search or an insertion in a B tree of order M with N items requires between $\log_M N$ and $\log_{M/2} N$ probes—a constant number, for practical purposes.*

Program 16.3 B-tree insertion

We insert new items by moving larger items to the right by one position, as in insertion sort. If the insertion overfills the node, we call `split` to divide the node into two halves, and return the link to the new node. One level up in the recursion, this extra link causes a similar insertion in the parent internal node, which could also split, possibly propagating the insertion all the way up to the root.

```
link insertR(link h, Item item, int H)
  { int i, j; Key v = key(item); entry x; link t, u;
    x.key = v; x.ref.item = item;
    if (H == 0)
      for (j = 0; j < h->m; j++)
        if (less(v, h->b[j].key)) break;
    if (H != 0)
      for (j = 0; j < h->m; j++)
        if ((j+1 == h->m) || less(v, h->b[j+1].key))
          {
            t = h->b[j++].ref.next;
            u = insertR(t, item, H-1);
            if (u == NULL) return NULL;
            x.key = u->b[0].key; x.ref.next = u;
            break;
          }
    for (i =(h->m)++; i > j; i--)
      h->b[i] = h->b[i-1];
    h->b[j] = x;
    if (h->m < M) return NULL; else return split(h);
  }
void STinsert(Item item)
  { link t, u = insertR(head, item, H);
    if (u == NULL) return;
    t = NEW(); t->m = 2;
    t->b[0].key = head->b[0].key;
    t->b[0].ref.next = head;
    t->b[1].key =  u->b[0].key;
    t->b[1].ref.next = u;
    head = t; H++;
  }
```

Program 16.4 B-tree node split

To split a node in a B tree, we create a new node, move the larger half of the keys to the new node, and then adjust counts and set sentinel keys in the middle of both nodes. This code assumes that `M` is even, and uses an extra position in each node for the item that causes the split. That is, the maximum number of keys in a node is `M-1`, and when a node gets `M` keys, we split it into two nodes with `M/2` keys each.

```
link split(link h)
  { int j; link t = NEW();
    for (j = 0; j < M/2; j++)
      t->b[j] = h->b[M/2+j];
    h->m = M/2; t->m = M/2;
    return t;
  }
```

This property follows from the observation that all the nodes in the interior of the B tree (nodes that are not the root and are not external) have between $M/2$ and M links, since they are formed from a split of a full node with M keys, and can only grow in size (when a lower node is split). In the best case, these nodes form a complete tree of degree M, which leads immediately to the stated bound (see Property 16.1). In the worst case, we have a complete tree of degree $M/2$. ■

When M is 1000, the height of the tree is less than three for N less than 125 million. In typical situations, we can reduce the cost to two probes by keeping the root node in internal memory. For a disk-searching implementation, we might take this step explicitly before embarking on any application involving a huge number of searches; in a virtual memory with caching, the root node will be the one most likely to be in fast memory, because it is the most frequently accessed node.

We can hardly expect to have a search implementation that can guarantee a cost of fewer than two probes for *search* and *insert* in huge files, and B trees are widely used because they allow us to achieve this ideal. The price of this speed and flexibility is the empty space within the nodes, which could be a liability for huge files.

Property 16.3 *A B tree of order M constructed from N random items is expected to have about $1.44N/M$ pages.*

Yao proved this fact in 1979, using mathematical analysis that is beyond the scope of this book (*see reference section*). It is based on analyzing a simple probabilistic model that describes tree growth. After the first $M/2$ nodes have been inserted, there are, at any given time, t_i external pages with i items, for $M/2 \leq i \leq M$, with $t_{M/2} + \ldots + t_M = N$. Since each interval between nodes is equally likely to receive a random key, the probability that a node with i items is hit is t_i/N. Specifically, for $i < M$, this quantity is the probability that the number of external pages with i items decreases by 1 and the number of external pages with $(i + 1)$ items increases by 1; and for $i = 2M$, this quantity is the probability that the number of external pages with $2M$ items decreases by 1 and the number of external pages with M items increases by 2. Such a probabilistic process is called a *Markov chain*. Yao's result is based on an analysis of the asymptotic properties of this chain. ■

We can also validate Property 16.3 by writing a program to simulate the probabilistic process (see Exercise 16.11 and Figures 16.8 and 16.9). Of course, we also could just build random B trees and measure their structural properties. The probabilistic simulation is simpler to do than either the mathematical analysis or the full implementation, and is an important tool for us to use in studying and comparing variants of the algorithm (see, for example, Exercise 16.16).

The implementations of other symbol-table operations are similar to those for several other tree-based representations that we have seen before, and are left as exercises (see Exercises 16.22 through 16.25). In particular, *select* and *sort* implementations are straightforward, but as usual, implementing a proper *delete* can be a challenging task. Like *insert*, most *delete* operations are a simple matter of removing an item from an external page and decrementing its counter, but what do we do when we have to remove an item from a node that has only $M/2$ items? The natural approach is to find items from sibling nodes to fill the space (perhaps reducing the number of nodes by one), but the implementation becomes complicated because we have to track down the keys associated with any items that we move among nodes. In practical situations, we can typically adopt the much simpler approach of letting external nodes become underfull, without suffering much performance degradation (see Exercise 16.25).

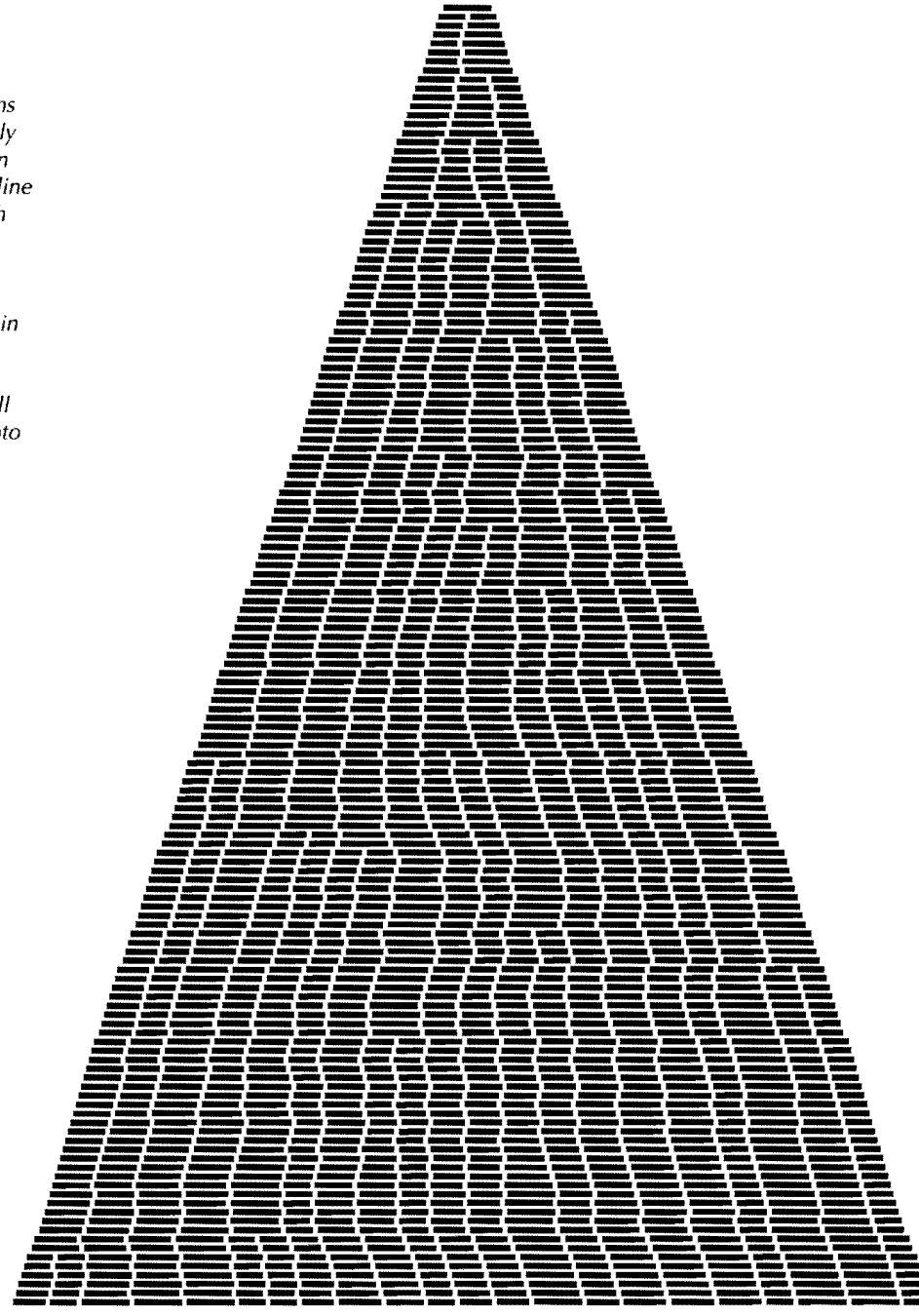


Figure 16.8
Growth of a large B tree

In this simulation, we insert items with random keys into an initially empty B tree with pages that can hold nine keys and links. Each line displays the external nodes, with each external node depicted as a line segment of length proportional to the number of items in that node. Most insertions land in an external node that is not full, increasing that node's size by 1. When an insertion lands in a full external node, the node splits into two nodes of half the size.

Many variations on the basic B-tree abstraction suggest themselves immediately. One class of variations saves time by packing as many page references as possible in internal nodes, thereby increasing the branching factor and flattening the tree. As we have discussed, the benefits of such changes are marginal in modern systems, since standard values of the parameters allow us to implement *search* and *insert* with two probes—an efficiency that we could hardly improve. Another class of variations improves storage efficiency by combining nodes with siblings before splitting. Exercises 16.13 through 16.16 are concerned with such a method, which reduces the excess storage used from 44 to 23 per cent, for random keys. As usual, the proper choice among different variations depends on properties of applications. Given the broad variety of different situations where B trees are applicable, we will not consider such issues in detail. We also will not be able to consider details of implementations, because there are so many device- and system-dependent matters to take into account. As usual, delving deeply into such implementations is a risky business, and we shy away from such fragile and nonportable code in modern systems, particularly when the basic algorithm performs so well.

Exercises

▷ **16.5** Give the contents of the 3-4-5-6 tree that results when you insert the keys E A S Y Q U E S T I O N W I T H P L E N T Y O F K E Y S in that order into an initially empty tree.

○ **16.6** Draw figures corresponding to Figures 16.5 through 16.7, to illustrate the process of inserting the keys 516, 177, 143, 632, 572, 161, 774, 470, 411, 706, 461, 612, 761, 474, 774, 635, 343, 461, 351, 430, 664, 127, 345, 171, and 357 in that order into an initially empty tree, with $M = 5$.

○ **16.7** Give the height of the B trees that result when you insert the keys in Exercise 16.28 in that order into an initially empty tree, for *each* value of $M > 2$.

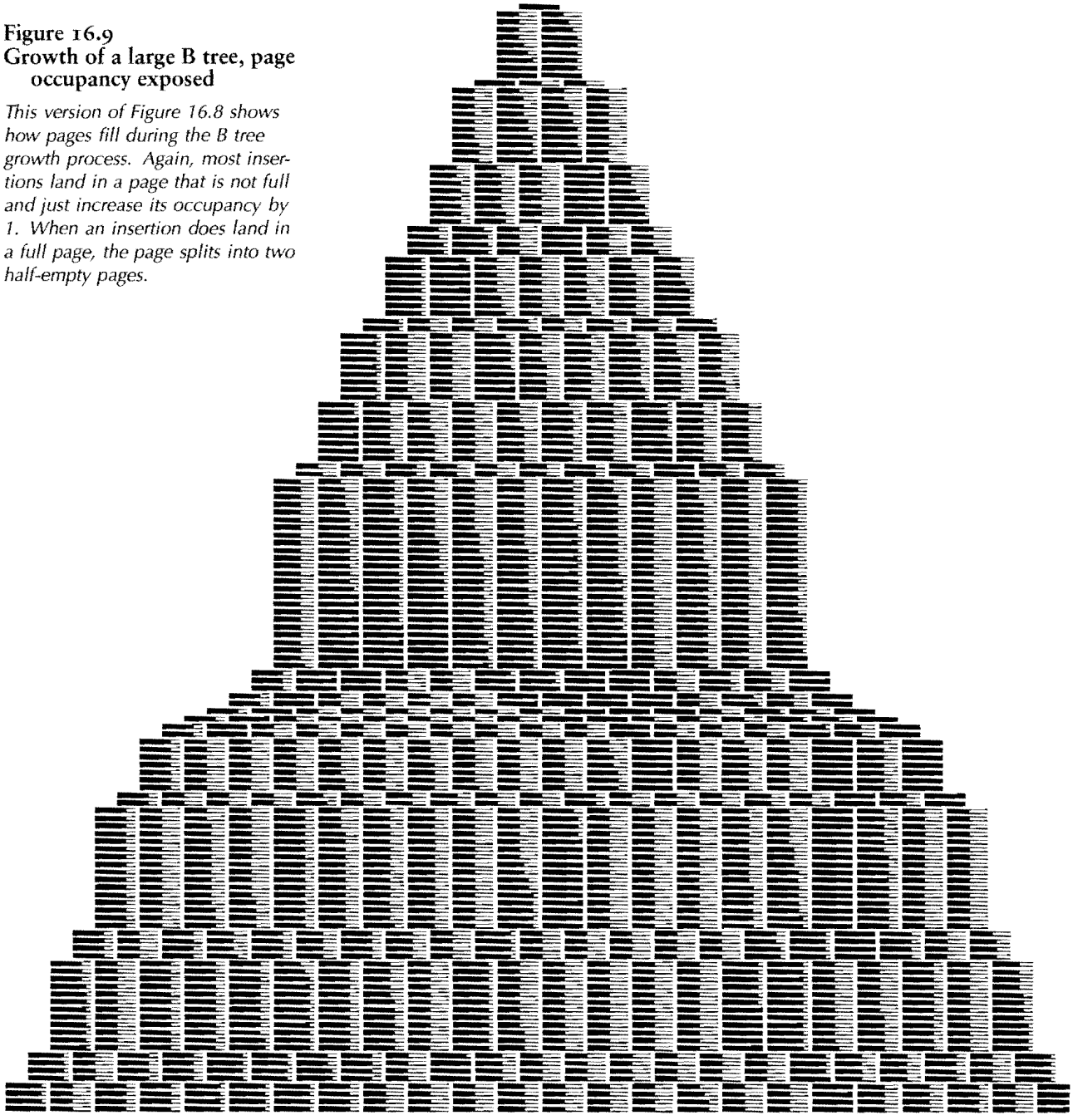
16.8 Draw the B tree that results when you insert 16 equal keys into an initially empty tree, with $M = 4$.

• **16.9** Draw the 1-2 tree that results when you insert the keys E A S Y Q U E S T I O N into an initially empty tree. Explain why 1-2 trees are not of practical interest as balanced trees.

• **16.10** Modify the B-tree–insertion implementation in Program 16.3 to do splitting on the way down the tree, in a manner similar to our implementation of 2-3-4–tree insertion (Program 13.6).

Figure 16.9
Growth of a large B tree, page occupancy exposed

This version of Figure 16.8 shows how pages fill during the B tree growth process. Again, most insertions land in a page that is not full and just increase its occupancy by 1. When an insertion does land in a full page, the page splits into two half-empty pages.



• **16.11** Write a program to compute the average number of external pages for a B tree of order M built from N random insertions into an initially empty tree, using the probabilistic process described after Property 16.1. Run your program for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

o **16.12** Suppose that, in a three-level tree, we can afford to keep a links in internal memory, between b and $2b$ links in pages representing internal nodes, and between c and $2c$ items in pages representing external nodes. What is the maximum number of items that we can hold in such a tree, as a function of a, b, and c?

o **16.13** Consider the *sibling split* (or *B* tree*) heuristic for B trees: When it comes time to split a node because it contains M entries, we combine the node with its sibling. If the sibling has k entries with $k < M - 1$, we reallocate the items giving the sibling and the full node each about $(M + k)/2$ entries. Otherwise, we create a new node and give each of the three nodes about $2M/3$ entries. Also, we allow the root to grow to hold about $4M/3$ items, splitting it and creating a new root node with two entries when it reaches that bound. State bounds on the number of probes used for a search or an insertion in a B* tree of order M with N items. Compare your bounds with the corresponding bounds for B trees (see Property 16.2), for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

•• **16.14** Develop a B* tree *insert* implementation (based on the sibling-split heuristic).

• **16.15** Create a figure like Figure 16.8 for the sibling-split heuristic.

• **16.16** Run a probabilistic simulation (see Exercise 16.11) to determine the average number of pages used when we use the sibling-split heuristic, building a B* tree of order M by inserting random nodes into an initially empty tree, for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

• **16.17** Write a program to construct a B tree index from the bottom up, starting with an array of pointers to pages containing between M and $2M$ items, in sorted order.

• **16.18** Could an index with all pages full, such as Figure 16.2, be constructed by the B-tree–insertion algorithm considered in the text (Program 16.3)? Explain your answer.

16.19 Suppose that many different computers have access to the same index, so several programs may be trying to insert a new node in the same B tree at about the same time. Explain why you might prefer to use top-down B trees instead of bottom-up B trees in such a situation. Assume that each program can (and does) delay the others from modifying any given node that it has read and might later modify.

• **16.20** Modify the B-tree implementation in Programs 16.1 through 16.3 to allow M items per node to exist in the tree.

▷ **16.21** Tabulate the difference between $\log_{999} N$ and $\log_{1000} N$, for $N = 10^3$, 10^4, 10^5, and 10^6.

▷ **16.22** Implement the *sort* operation for a B-tree–based symbol table.

○ **16.23** Implement the *select* operation for a B-tree–based symbol table.

●● **16.24** Implement the *delete* operation for a B-tree–based symbol table.

○ **16.25** Implement the *delete* operation for a B-tree–based symbol table, using a simple method where you delete the indicated item from its external page (perhaps allowing the number of items in the page to fall below $M/2$), but do not propagate the change up through the tree, except possibly to adjust the key values if the deleted item was the smallest in its page.

● **16.26** Modify Programs 16.2 and 16.3 to use binary search (see Program 12.6) within nodes. Determine the value of M that minimizes the time that your program takes to build a symbol table by inserting N items with random keys into an initially empty table, for $N = 10^3$, 10^4, 10^5, and 10^6, and compare the times that you get with the corresponding times for red–black trees (Program 13.6).

16.4 Extendible Hashing

An alternative to B trees that extends digital searching algorithms to apply to external searching was developed in 1978 by Fagin, Nievergelt, Pippenger, and Strong. Their method, called *extendible hashing*, leads to a *search* implementation that requires just one or two probes for typical applications. The corresponding *insert* implementation also (almost always) requires just one or two probes.

Extendible hashing combines features of hashing, multiway-trie algorithms, and sequential-access methods. Like the hashing methods of Chapter 14, extendible hashing is a randomized algorithm—the first step is to define a hash function that transforms keys into integers (see Section 14.1). For simplicity, in this section, we simply consider keys to be random fixed-length bitstrings. Like the multiway-trie algorithms of Chapter 15, extendible hashing begins a search by using the leading bits of the keys to index into a table whose size is a power of 2. Like B-tree algorithms, extendible hashing stores items on pages that are split into two pieces when they fill up. Like indexed sequential-access methods, extendible hashing maintains a directory that tells us where we can find the page containing the items that match the search key. The blending of these familiar features in one algorithm makes extendible hashing a fitting conclusion to our study of search algorithms.

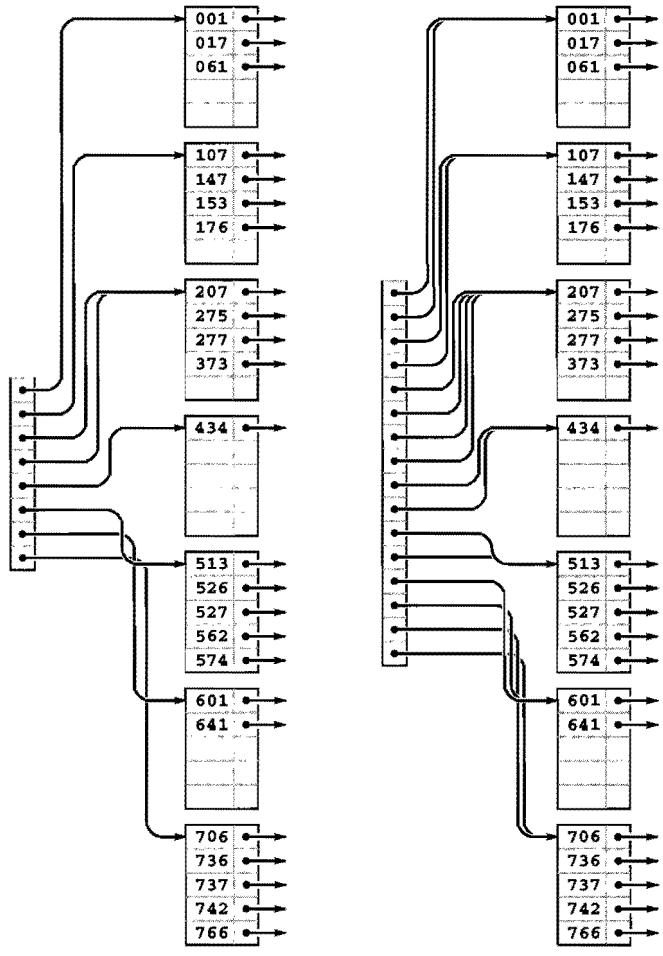


Figure 16.10
Directory page indices

With a directory of eight entries, we can store up to 40 keys by storing all records whose first 3 bits match on the same page, which we can access via a pointer stored in the directory (left). *Directory entry 0 contains a pointer to the page that contains all keys that begin with 000; table entry 1 contains a pointer to the page that contains all keys that begin with 001; table entry 2 contains a pointer to the page that contains all keys that begin with 010, and so forth. If some pages are not fully populated, we can reduce the number of pages required by having multiple directory pointers to a page. In this example* (left), 373 *is on the same page as the keys that start with* 2; *that page is defined to be the page that contains items with keys whose first 2 bits are* 01.

If we double the size of the directory and clone each pointer, we get a structure that we can index with the first 4 bits of the search key (right). *For example, the final page is still defined to be the page that contains items with keys whose first three bits are* 111, *and it will be accessed through the directory if the first 4 bits of the search key are* 1110 *or* 1111. *This larger directory can accommodate growth in the table.*

Suppose that the number of disk pages that we have available is a power of 2—say 2^d. Then, we can maintain a directory of the 2^d different page references, use d bits of the keys to index into the directory, and can keep, on the same page, all keys that match in their first k bits, as illustrated in Figure 16.10. As we do with B trees, we keep the items in order on the pages, and do sequential search once we reach the page corresponding to an item with a given search key.

Figure 16.10 illustrates the two basic concepts behind extendible hashing. First, we do not necessarily need to maintain 2^d pages. That is, we can arrange to have multiple directory entries refer to the same page, without changing our ability to search the structure quickly, by

Program 16.5 Extendible hashing definitions and initialization

An extendible hash table is a directory of references to pages (like the external nodes in B trees) that contain up to $2M$ items. Each page also contains a count (m) of the number of items on the page, and an integer (k) that specifies the number of leading bits for which we know the keys of the items to be identical. As usual, N specifies the number of items in the table. The variable d specifies the number of bits that we use to index into the directory, and D is the number of directory entries, so $D = 2^d$. The table is initially set to a directory of size 1, which points to an empty page.

```
typedef struct STnode* link;
struct STnode { Item b[M]; int m; int k; };
static link *dir;
static int d, D, N;
link NEW()
  { link x = malloc(sizeof *x);
    x->m = 0;  x->k = 0;
    return x;
  }
void STinit(int maxN)
  {
    d = 0; N = 0; D = 1;
    dir = malloc(D*(sizeof *dir));
    dir[0] = NEW();
  }
```

combining keys with differing values for their leading d bits together on the same page, while still maintaining our ability to find the page containing a given key by using the leading bits of the key to index into the directory. Second, we can double the size of the directory to increase the capacity of the table.

Specifically, the data structure that we use for extendible hashing is much simpler than the one that we used for B trees. It consists of pages that contain up to M items, and a directory of 2^d pointers to pages (see Program 16.5). The pointer in directory location x refers to the page that contains all items whose leading d bits are equal to x. The table is constructed with d sufficiently large that we are guaranteed that there are less than M items on each page. The implementation

Program 16.6 Extendible hashing search

Searching in an extendible hashing table is simply a matter of using the leading bits of the key to index into the directory, then doing a sequential search on the specified page for an item with a key equal to the search key. The only requirement is that each directory entry refer to a page that is guaranteed to contain all items in the symbol table that begin with the specified bits.

```
Item search(link h, Key v)
  { int j;
    for (j = 0; j < h->m; j++)
      if (eq(v, key(h->b[j])))
        return h->b[j];
    return NULLitem;
  }
Item STsearch(Key v)
  { return search(dir[bits(v, 0, d)], v); }
```

of *search* is simple: We use the leading d bits of the key to index into the directory, which gives us access to the page that contains any items with matching keys, then do sequential search for such an item on that page (see Program 16.6).

The data structure needs to become slightly more complicated to support *insert*, but one of its essential features is that this search algorithm works properly without any modification. To support *insert*, we need to address the following questions:

- What do we do when the number of items that belong on a page exceeds that page's capacity?
- What directory size should we use?

For example, we could not use $d = 2$ in the example in Figure 16.10 because some pages would overflow, and we would not use $d = 5$ because too many pages would be empty. As usual, we are most interested in supporting the *insert* operation for the symbol-table ADT, so that, for example, the structure can grow gradually as we do a series of intermixed *search* and *insert* operations. Taking this point of view corresponds to refining our first question:

- What do we do when we need to *insert* an item into a full page?

For example, we could not insert an item whose key starts with a 5 or a 7 in the example in Figure 16.10 because the corresponding pages are full.

Definition 16.3 *An* **extendible hash table** *of order d is a directory of 2^d references to pages that contain up to M items with keys. The items on each page are identical in their first k bits, and the directory contains 2^{d-k} pointers to the page, starting at the location specified by the leading k bits in the keys on the page.*

Some d-bit patterns may not appear in any keys. We leave the corresponding directory entries unspecified in Definition 16.3, although there is a natural way to organize pointers to null pages; we will examine it shortly.

To maintain these characteristics as the table grows, we use two basic operations: a *page split*, where we distribute some of the keys from a full page onto another page; and a *directory split*, where we double the size of the directory and increase d by 1. Specifically, when a page fills, we split it into two pages, using the leftmost bit position for which the keys differ to decide which items go to the new page. When a page splits, we adjust the directory pointers appropriately, doubling the size of the directory if necessary.

As usual, the best way to understand the algorithm is to trace through its operation as we insert a set of keys into an initially empty table. Each of the situations that the algorithm must address occurs early in the process, in a simple form, and we soon come to a realization of the algorithm's underlying principles. Figures 16.11 through 16.13 show the construction of an extendible hash table for the sample set of 25 octal keys that we have been considering in this chapter. As occurs in B trees, most of the insertions are uneventful: They simply add a key to a page. Since we start with one page and end up with eight pages, we can infer that seven of the insertions caused a page split; since we start with a directory of size 1 and end up with a directory of size 16, we can infer that four of the insertions caused a directory split.

Property 16.4 *The extendible hash table built from a set of keys depends on only the values of those keys, and does not depend on the order in which the keys are inserted.*

Consider the trie corresponding to the keys (see Property 15.2), with each internal node labeled with the number of items in its subtree. An

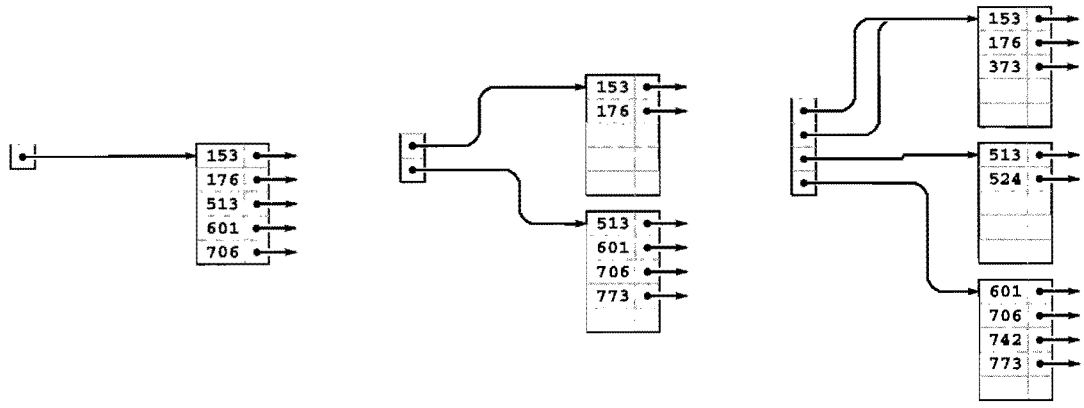


Figure 16.11
Extendible hash table construction, part 1

As in B trees, the first five insertions into an extendible hash table go into a single page (left). *Then, when we insert* 773, *we split into two pages (one with all the keys beginning with a 0 bit and one with all the keys beginning with a 1 bit) and double the size of the directory to hold one pointer to each of the pages* (center). *We insert* 742 *into the bottom page (because it begins with a 1 bit) and* 373 *into the top page (because it begins with a 0 bit), but we then need to split the bottom page to accommodate* 524. *For this split, we put all the items with keys that begin with* 10 *on one page and all the items with keys that begin with* 11 *on the other, and we again double the size of the directory to accommodate pointers to both of these pages* (right). *The directory contains two pointers to the page containing items with keys starting with a 0 bit: one for keys that begin with* 00 *and the other for keys that begin with* 01.

internal node corresponds to a page in the extendible hash table if and only if its label is less than M and its parent's label is not less than M. All the items below the node go on that page. If a node is at level k, it corresponds to a k-bit pattern derived from the trie path in the normal way, and all entries in the extendible hash table's directory with indices that begin with that k-bit pattern contain pointers to the corresponding page. The size of the directory is determined by the deepest level among all the internal nodes in the trie that correspond to pages. Thus, we can convert a trie to an extendible hash table without regard to the order in which items are inserted, and this property holds as a consequence of Property 15.2. ■

Program 16.7 is an implementation of the *insert* operation for an extendible hash table. First, we access the page that could contain the search key, with a single reference to the directory, as we did for search. Then, we insert the new item there, as we did for external nodes in B trees (see Program 16.2). If this insertion leaves M items in the node, then we invoke a split function, again as we did for B trees, but the split function is more complicated in this case. Each page contains the number k of leading bits that we know to be the same in the keys of all the items on the page, and, because we number bits from the left starting at 0, k also specifies the index of the bit that we want to test to determine how to split the items.

Therefore, to split a page, we make a new page, then put all the items for which that bit is 0 on the old page and all the items for which that bit is 1 on the new page, then set the bit count to $k + 1$ for

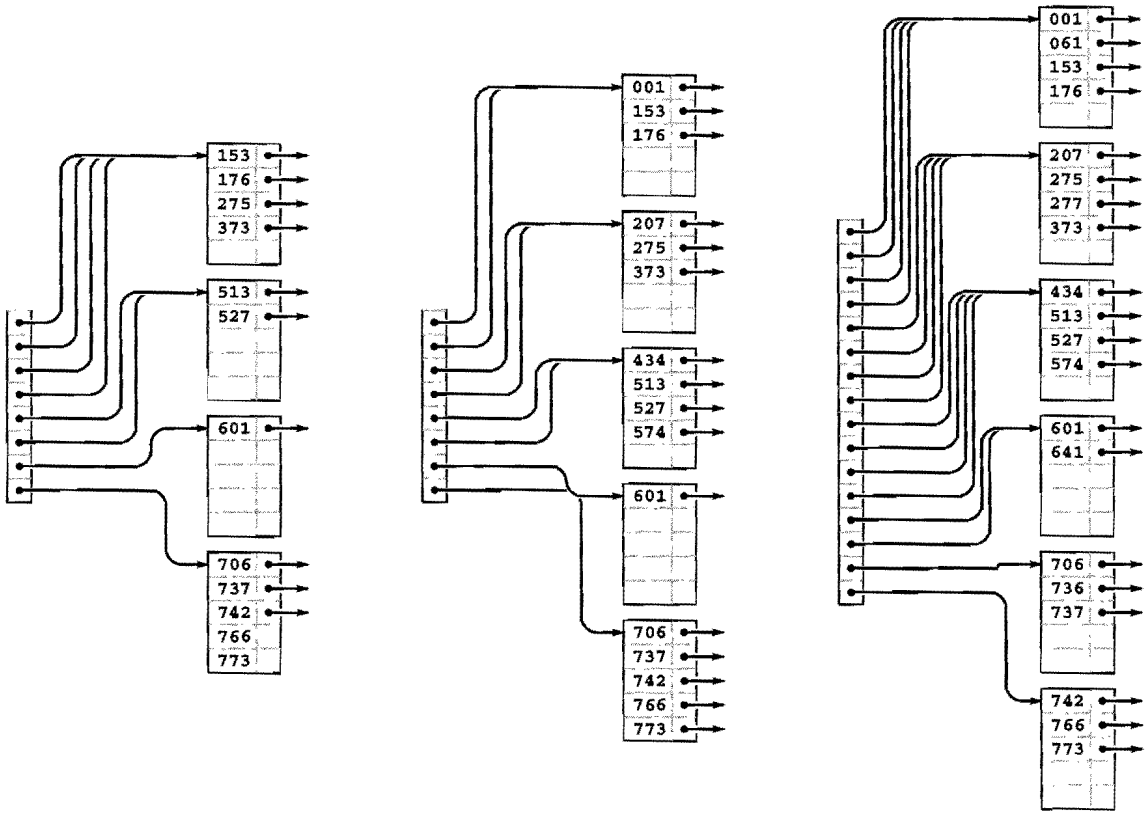


Figure 16.12
Extendible hash table construction, part 2

We insert the keys 766 *and* 275 *into the rightmost B tree in Figure 16.11 without any node splits* (left). *Then, when we insert* 737, *the bottom page splits, and that, because there is only one link to the bottom page, causes a directory split* (center). *Then, we insert* 574, 434, 641, *and* 207 *before* 001 *causes the top page to split* (right).

both pages. Now, it could be the case that all the keys have the same value for bit k, which would still leave us with a full node. If so, we simply go on to the next bit, continuing until we get a least one item in each page. The process must terminate, eventually, *unless we have M values of the same key*. We discuss that case shortly.

As with B trees, we leave space for an extra entry in every page to allow splitting after insertion, thus simplifying the code. Again, this technique has little practical effect, and we can ignore the effect in the analysis.

When we create a new page, we have to insert a pointer to it in the directory. The code that accomplishes this insertion is given in Program 16.8. The simplest case to consider is the one where the directory, prior to insertion, has precisely two pointers to the page that splits. In that case, we need simply to arrange to set the second pointer

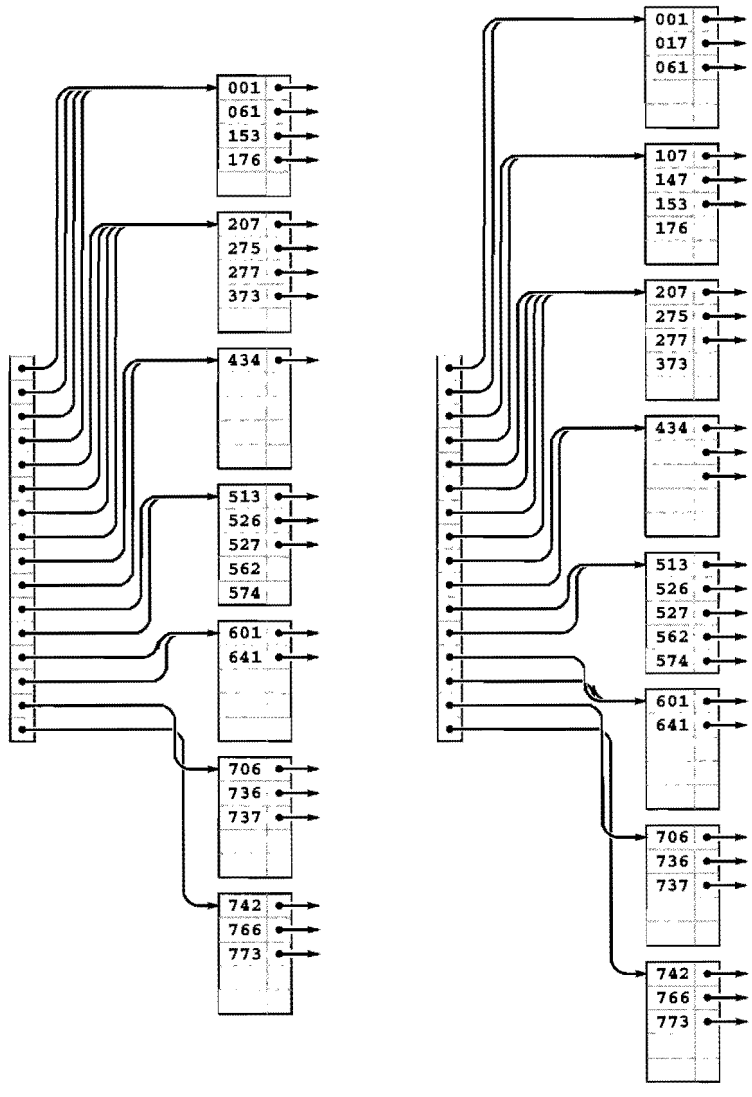


Figure 16.13
Extendible hash table construction, part 3

Continuing the example in Figures 16.11 and 16.12, we insert the 5 keys 526, 562, 017, 107, *and* 147 *into the rightmost B tree in Figure 16.6. Node splits occur when we insert* 526 (left) *and* 107 (right).

Program 16.7 Extendible hashing insertion

To insert an item into an extendible hash table, we search; then we insert the item on the specified page; then we split the page if the insertion caused overflow. The general scheme is the same as that for B trees, but the methods that we use to find the appropriate page and to split pages are different.

The split function creates a new node, then examines the kth bit (counting from the left) of each item's key: if the bit is 0, the item stays in the old node; if it is 1, it goes in the new node. The value $k + 1$ is assigned to the "leading bits known to be identical" field of both nodes after the split. If this process does not result in at least one key in each node, we split again, until the items are so separated. At the end, we insert the pointer with the new node into the directory.

```
link split(link h)
  { int j; link t = NEW();
    while (h->m == 0 || h->m == M)
      {
        h->m = 0; t->m = 0;
        for (j = 0; j < M; j++)
          if (bits(h->b[j], h->k, 1) == 0)
               h->b[(h->m)++] = h->b[j];
          else t->b[(t->m)++] = h->b[j];
        t->k = ++(h->k);
      }
    insertDIR(t, t->k);
  }
void insert(link h, Item item)
  { int i, j; Key v = key(item);
    for (j = 0; j < h->m; j++)
      if (less(v, key(h->b[j]))) break;
    for (i = (h->m)++; i > j; i--)
      h->b[i] = h->b[i-1];
    h->b[j] = item;
    if (h->m == M) split(h);
  }
void STinsert(Item item)
  { insert(dir[bits(key(item), 0, d)], item); }
```

Program 16.8 Extendible-hashing directory insertion

This deceptively simple code is at the heart of the extendible-hashing process. We are given a link `t` to a node that carries items that match in the first k bits, which is to be incorporated into the directory. In the simplest case, where `d` and `k` are equal, we just put `t` into `d[x]`, where `x` is the value of the first `d` bits of `t->b[0]` (and of all the other items on the page). If `k` is greater than `d`, we have to double the size of the directory, until reducing to the case where `d` and `k` are equal. If `k` is less than `d`, we need to set more than one pointer—the first `for` loop calculates the number of pointers that we need to set (2^{d-k}), and the second `for` loop does the job.

```
void insertDIR(link t, int k)
  { int i, m, x = bits(t->b[0], 0, k);
    while (d < k)
      { link *old = dir;
        d += 1; D += D;
        dir = malloc(D*(sizeof *dir));
        for (i = 0; i < D; i++) dir[i] = old[i/2];
        if (d < k) dir(bits(x, 0, d) ^ 1) = NEW();
      }
    for (m = 1; k < d; k++) m *= 2;
    for (i = 0; i < m; i++) dir[x*m+i] = t;
  }
```

to reference the new page. If the number of bits k that we need to distinguish the keys on the new page is greater than the number of bits d that we have to access the directory, then we have to increase the size of the directory to accommodate the new entry. Finally, we update the directory pointers as appropriate.

If more than M items have duplicate keys, the table overflows, and the code in Program 16.7 goes into an infinite loop, looking for a way to distinguish the keys. A related problem is that the directory may get unnecessarily huge, if the keys have an excessive number of leading bits that are equal. This situation is akin to the excessive time required for MSD radix sort, for files that have large numbers of duplicate keys or long stretches of bit positions where they are identical. We depend on the randomization provided by the hash function to stave off these problems (see Exercise 16.43). Even with hashing, extraordinary steps

must be taken if large numbers of duplicate keys are present, because hash functions take equal keys to equal hash values. Duplicate keys can make the directory artificially large; and the algorithm breaks down entirely if there are more equal keys than fit in one page. Therefore, we need to add tests to guard against the occurrence of these conditions before using this code (see Exercise 16.35).

The primary performance parameters of interest are the number of pages used (as with B trees) and the size of the directory. The randomization for this algorithm is provided by the hash functions, so average-case performance results apply to any sequence of N distinct insertions.

Property 16.5 *With pages that can hold M items, extendible hashing requires about $1.44(N/M)$ pages for a file of N items, on the average. The expected number of entries in the directory is about $3.92(N^{1/M})(N/M)$.*

This (rather deep) result extends the analysis of tries that we discussed briefly in the previous chapter (*see reference section*). The exact constants are $\lg e = 1/\ln 2$ for the number of pages and $e \lg e = e/\ln 2$ for the directory size, though the precise values of the quantities oscillate around these average values. We should not be surprised by this phenomenon because, for example, the directory size has to be a power of 2, a fact which has to be accounted for in the result. ▪

Note that the growth rate of the directory size is faster than linear in N, particularly for small M. However, for N and M in ranges of practical interest, $N^{1/M}$ is quite close to 1, so we can expect the directory to have about $4(N/M)$ entries, in practice.

We have considered the directory to be a single array of pointers. We can keep the directory in memory, or, if it is too big, we can keep a root node in memory that tells where the directory pages are, using the same indexing scheme. Alternatively, we can add another level, indexing the first level on the first 10 bits (say), and the second level on the rest of the bits (see Exercise 16.36).

As we did for B trees, we leave the implementation of other symbol-table operations for exercises (see Exercises 16.38 and 16.41). Also as it is with B trees, a proper *delete* implementation is a challenge, but allowing underfull pages is an easy alternative that can be effective in many practical situations.

Exercises

▷ 16.27 How many pages would be empty if we were to use a directory of size 32 in Figure 16.10?

16.28 Draw figures corresponding to Figures 16.11 through 16.13, to illustrate the process of inserting the keys 562, 221, 240, 771, 274, 233, 401, 273, and 201 in that order into an initially empty tree, with $M = 5$.

○ 16.29 Draw figures corresponding to Figures 16.11 through 16.13, to illustrate the process of inserting the keys 562, 221, 240, 771, 274, 233, 401, 273, and 201 in that order into an initially empty tree, with $M = 5$.

○ 16.30 Assume that you are given an array of items in sorted order. Describe how you would determine the directory size of the extendible hash table corresponding to that set of items.

• 16.31 Write a program that constructs an extendible hash table from an array of items that is in sorted order, by doing two passes through the items: one to determine the size of the directory (see Exercise 16.30) and one to allocate the items to pages and fill in the directory.

○ 16.32 Give a set of keys for which the corresponding extendible hash table has directory size 16, with eight pointers to a single page.

•• 16.33 Create a figure like Figure 16.8 for extendible hashing.

• 16.34 Write a program to compute the average number of external pages and the average directory size for an extendible hash table built from N random insertions into an initially empty tree, when the page capacity is M. Compute the percentage of empty space, for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

16.35 Add appropriate tests to Program 16.7 to guard against malfunction in case too many duplicate keys or keys with too many leading equal bits are inserted into the table.

• 16.36 Modify the extendible-hashing implementation in Programs 16.5 through 16.7 to use a two-level directory, with no more than M pointers per directory node. Pay particular attention to deciding what to do when the directory first grows from one level to two.

• 16.37 Modify the extendible-hashing implementation in Programs 16.5 through 16.7 to allow M items per page to exist in the data structure.

○ 16.38 Implement the *sort* operation for an extendible hash table.

○ 16.39 Implement the *select* operation for an extendible hash table.

•• 16.40 Implement the *delete* operation for an extendible hash table.

○ 16.41 Implement the *delete* operation for an extendible hash table, using the method indicated in Exercise 16.25.

•• 16.42 Develop a version of extendible hashing that splits pages when splitting the directory, so that each directory pointer points to a unique page. Develop experiments to compare the performance of your implementation to that of the standard implementation.

o 16.43 Run empirical studies to determine the number of random numbers that we would expect to generate before finding more than M numbers with the same d initial bits, for $M = 10$, 100, and 1000, and for $1 \leq d \leq 20$.

• 16.44 Modify hashing with separate chaining (Program 14.3) to use a hash table of size $2M$, and keep items in pages of size $2M$. That is, when a page fills, link it to a new empty page, so each hash table entry points to a linked list of pages. Empirically determine the average number of probes required for a search after building a table from N items with random keys, for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6.

o 16.45 Modify double hashing (Program 14.6) to use pages of size $2M$, treating accesses to full pages as "collisions." Empirically determine the average number of probes required for a search after building a table from N items with random keys, for $M = 10$, 100, and 1000 and $N = 10^3$, 10^4, 10^5, and 10^6, using an initial table size of $3N/2M$.

o 16.46 Develop a symbol-table implementation using extendible hashing that supports the *initialize*, *count*, *search*, *insert*, *delete*, *join*, *select*, and *sort* operations for first-class symbol-table ADTs with client item handles (see Exercises 12.4 and 12.5).

16.5 Perspective

The most important application of the methods discussed in this chapter is to construct indexes for huge databases that are maintained on external memory—for example, in disk files. Although the underlying algorithms that we have discussed are powerful, developing a file-system implementation based on B trees or on extendible hashing is a complex task. First, we cannot use the C programs in this section directly—they have to be modified to read and refer to disk files. Second, we have to be sure that the algorithm parameters (page and directory size, for example) are tuned properly to the characteristics of the particular hardware that we are using. Third, we have to pay attention to reliability, and to error detection and correction. For example, we need to be able to check that the data structure is in a consistent state, and to consider how we might proceed to correct any of the scores of errors that might crop up. Systems considerations of this kind are critical—and are beyond the scope of this book.

On the other hand, if we have a programming system that supports virtual memory, we can put to direct use the C implementations that we have considered here in a situation where we have a huge number of symbol-table operations to perform on a huge table. Roughly, each time that we access a page, such a system will put that page in a *cache*, where references to data on that page are handled efficiently. If we refer to a page that is not in the cache, the system has to read the page from external memory, so we can think of cache misses as roughly equivalent to the probe cost measure that we have been using.

For B trees, every search or insertion references the root, so the root will always be in the cache. Otherwise, for sufficiently large M, typical searches and insertions involve at most two cache misses. For a large cache, there is a good chance that the first page (the child of the root) that is accessed on a search is already in the cache, so the average cost per search is likely to be significantly less than two probes.

For extendible hashing, it is unlikely that the whole directory will be in the cache, so we expect that both the directory access and the page access might involve a cache miss (this case is the worst case). That is, two probes are required for a search in a huge table, one to access the appropriate part of the directory and one to access the appropriate page.

These algorithms form an appropriate subject on which to close our discussion of searching, because, to use them effectively, we need to understand basic properties of binary search, BSTs, balanced trees, hashing, and tries—the basic searching algorithms that we have studied in Chapters 12 through 15. As a group, these algorithms provide us with solutions to the symbol-table implementation problem in a broad variety of situations: they constitute an outstanding example of the power of algorithmic technology.

Exercises

16.47 Modify the B-tree implementation in Section 16.3 (Programs 16.1 through 16.3) to use an ADT for page references.

16.48 Modify the extendible-hashing implementation in Section 16.4 (Programs 16.5 through 16.8) to use an ADT for page references.

16.49 Estimate the average number of probes per search in a B tree for S random searches, in a typical cache system, where the T most-recently-accessed pages are kept in memory (and therefore add 0 to the probe count). Assume that S is much larger than T.

16.50 Estimate the average number of probes per search in an extendible hash table, for the cache model described in Exercise 16.49.

o 16.51 If your system supports virtual memory, design and conduct experiments to compare the performance of B trees with that of binary search, for random searches in a huge symbol table.

16.52 Implement a priority-queue ADT that supports *construct* for a huge number of items, followed by a huge number of *insert* and *delete the maximum* operations (see Chapter 9).

16.53 Develop an external symbol-table ADT based on a skip-list representation of B trees (see Exercise 13.80).

• 16.54 If your system supports virtual memory, run experiments to determine the value of M that leads to the fastest search times for a B tree implementation supporting random *search* operations in a huge symbol table. (It may be worthwhile for you to learn basic properties of your system before conducting such experiments, which can be costly.)

•• 16.55 Modify the B-tree implementation in Section 16.3 (Programs 16.1 through 16.3) to operate in an environment where the table resides on external storage. If your system allows nonsequential file access, put the whole table on a single (huge) file, and use offsets within the file in place of pointers in the data structure. If your system allows you to access pages on external devices directly, use page addresses in place of pointers in the data structure. If your system allows both, choose the approach that you determine to be most reasonable for implementing a huge symbol table.

•• 16.56 Modify the extendible-hashing implementation in Section 16.4 (Programs 16.5 through 16.8) to operate in an environment where the table resides on external storage. Explain the reasons for the approach that you choose for allocating the directory and the pages to files (see Exercise 16.55).

References for Part Four

The primary references for this section are the books by Knuth; Baeza-Yates and Gonnet; Mehlhorn; and Cormen, Leiserson, and Rivest. Many of the algorithms covered here are treated in great detail in these books, with mathematical analyses and suggestions for practical applications. Classical methods are covered thoroughly in Knuth; the more recent methods are described in the other books, with further references to the literature. These four sources, and the Sedgewick-Flajolet book, describe nearly all the "beyond the scope of this book" material referred to in this section.

The material in Chapter 13 comes from the 1996 paper by Roura and Martinez, the 1985 paper by Sleator and Tarjan, and the 1978 paper by Guibas and Sedgewick. As suggested by the dates of these papers, balanced trees are the subject of ongoing research. The books cited above have detailed proofs of properties of red–black trees and similar structures, and references to more recent work.

The treatment of tries in Chapter 15 is classical (though C implementations are rarely found in the literature). The material on TSTs comes from the 1997 paper by Bentley and Sedgewick.

The 1972 paper by Bayer and McCreight introduced B trees, and the extendible hashing algorithm presented in Chapter 16 comes from the 1979 paper by Fagin, Nievergelt, Pippenger, and Strong. Analytic results on extendible hashing were derived by Flajolet in 1983. These papers are must reading for anyone wishing further information on external searching methods. Practical applications of these methods arise within the context of database systems. An introduction to this field is given, for example, in the book by Date.

R. Baeza-Yates and G. H. Gonnet, *Handbook of Algorithms and Data Structures*, second edition, Addison-Wesley, Reading, MA, 1984.

J. L. Bentley and R. Sedgewick, "Sorting and searching strings," Eighth Symposium on Discrete Algorithms, New Orleans, January, 1997.

R. Bayer and E. M. McCreight, "Organization and maintenance of large ordered indexes," *Acta Informatica* 1, 1972.

T. H. Cormen, C. E. Leiserson, and R. L. Rivest, *Introduction to Algorithms*, MIT Press, 1990.

C. J. Date, *An Introduction to Database Systems*, sixth edition, Addison-Wesley, Reading, MA, 1995.

R. Fagin, J. Nievergelt, N. Pippenger, and H. R. Strong, "Extendible hashing—a fast access method for dynamic files," *ACM Transactions on Database Systems* 4, 1979.

P. Flajolet, "On the performance analysis of extendible hashing and trie search," *Acta Informatica* 20, 1983.

L. Guibas and R. Sedgewick, "A dichromatic framework for balanced trees," in *19th Annual Symposium on Foundations of Computer Science*, IEEE, 1978. Also in *A Decade of Progress 1970–1980*, Xerox PARC, Palo Alto, CA.

D. E. Knuth, *The Art of Computer Programming. Volume 3: Sorting and Searching*, second edition, Addison-Wesley, Reading, MA, 1997.

K. Mehlhorn, *Data Structures and Algorithms 1: Sorting and Searching*, Springer-Verlag, Berlin, 1984.

S. Roura and C. Martinez, "Randomization of search trees by subtree size," Fourth European Symposium on Algorithms, Barcelona, September, 1996.

R. Sedgewick and P. Flajolet, *An Introduction to the Analysis of Algorithms*, Addison-Wesley, Reading, MA, 1996.

D. Sleator and R. E. Tarjan, "Self-adjusting binary search trees," *Journal of the ACM* 32, 1985.

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